

Finite-Prefix Corrections and Dual Certificates for Polyomino Growth

An exact bound $\lambda \leq 4.498$
and a certificate theorem for convolution recurrences

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Abstract

We develop two extensions of the seventeen-neighborhood polyomino project in ProveIt. First, exact small-size occurrence counts are incorporated into a positive convolution system by subtracting its finite coefficient defect. A nonnegative tail formulation makes this correction rigorous without assuming convergence of any generating function. Enumerating the seventeen marked neighborhoods through size 18 gives an exact rational certificate for

$$\lambda \leq \frac{2249}{500} = 4.498 < 4.5,$$

improving the repository's 4.5235 upper bound. Second, balanced integer weights on monomials produce finite arithmetic certificates of divergence. We prove completeness for productive, strongly connected, nonlinear positive polynomial systems, thereby providing a certificate-based answer to Bui's Question 2 for this class. An explicit 53-weight certificate and the repository's primal certificate bracket the uncorrected recurrence growth between 4.52349 and 4.5235. We also prove a spectral stability criterion for the prefix hierarchy and give a counterexample in which arbitrarily long exact prefixes still fail to recover the true exponential growth. Full proofs, exact data, enumerators, rational verifiers, and proposed research directions are supplied.

Status and scope. The new bound is a computer-assisted mathematical result, not a newly kernel-checked Lean theorem. The new arguments are proved in this report; the finite enumeration and arithmetic are reproducible. The geometric recurrence lemma is imported from Bui and the pinned repository. The work has not been independently refereed. General geometric-programming duality and polyomino enumeration algorithms are prior tools, not claimed inventions of this report.

[write] In ProveIt the geometric recurrence lemma is more than imported: Lean proves Bui's Lemma 4 recurrences for the actual marked counts (`geometricPublishedBuiRecurrences`), and the project's Lean-verified bound is $\lambda \leq 9047/2000 = 4.5235$. The bound $\lambda \leq 2249/500 = 4.498$ of this report is computer-assisted and is *not* formalized. See [section 1.5](#).

[Added 29 September 2026, batch 40:] Part II ([section 12](#)), also unformalized, proves that no exact prefix takes the fixed map's majorant growth below 4.3149: a limit of the method, *not* a lower bound on λ and *not* an improvement on 4.498. *Tempting false readings:* " $\lambda \geq 4.3149$ "; " $4.3149 < 4.498$ is a better bound". Both are false.

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1 Problem selection, sources, and main contributions

1.1 Why this repository thread

The ProveIt repository contains a concrete project on fixed square-lattice polyominoes at

Combinatorics/Polyominoes/KlarnerConstant/.

Its existing endpoint is an upper bound of $9047/2000 = 4.5235$ for Klarner’s constant. The relevant sources were read at commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

In particular, the project README, the independent rational verifier, Patterns.lean, and GeometricComplete.lean fix the mathematical and computational interface used here [2, 3, 4, 5, 6]. The last of these files assembles the seventeen geometric recurrences for actual marked polyomino counts and then applies the 4.5235 certificate. We inspected those source statements; we did not run a fresh Lean build.

The published input is Vuong Bui’s *A convolutional approach to bounding the number of polyominoes*, arXiv:2511.00461v2, dated 6 May 2026 [1]. Its Lemma 4 and Appendix B give the seventeen-neighborhood recurrence system. Its Section 3, Question 2 asks for compact, readily verifiable lower-growth certificates for convolution recurrences. This is a particularly suitable target: the geometric inequalities are already available, the numerical improvement can have an exact proof boundary, and the certificate question has a precise published formulation.

[write] Earlier upper bounds. The delivered text compares its bound only with Bui’s 4.5238 and the repository’s 4.5235. For context, the repository’s research note on the 4.5235 bound [7] records the upper-bound history as follows: Eden obtained 6.75 [8]; Klarner and Rivest successively obtained 5, 4.83 and 4.649551, the last by the procedure of [9]; Barequet and Shalah reached 4.5252 using over twenty-three trillion local configurations [10]; Bui’s seventeen neighborhood types then gave 4.5238 [1], and the repository’s exact supersolution for Bui’s unchanged system gave $9047/2000 = 4.5235$, which is kernel-checked in Lean (section 1.5). The bound 4.498 of theorem 4.1 continues this list, but it is computer-assisted and not formalized. None of the earlier methods used exact marked prefixes of the kind introduced here; this is a statement about the cited sources, not a priority search.

1.2 What is proved here

The first result concerns actual polyominoes. If A_n is the number of translation classes of n -cell polyominoes, then the data and certificate in this report give

$$A_n \leq \frac{927884613}{10^9} \left(\frac{2249}{500} \right)^n \quad (n \geq 1), \quad \lambda \leq \frac{2249}{500}. \quad (1)$$

The factor $927884613/10^9$ is the G coordinate of the supplied rational supersolution. No numerical eigenvalue or floating-point root appears in the verification of (1).

The second result concerns the equality recurrences used as majorants. We give a theorem of alternatives: under explicit productivity, irreducibility, and nonlinearity hypotheses, either a positive supersolution exists or a balanced integer-monomial certificate disproves its existence. The two alternatives cannot occur together. This supplies the requested type of lower-growth certificate for a class containing Bui’s seventeen-variable system. The explicit example proves

$$4.52349 \leq \mu_{\text{original}} \leq 4.5235. \quad (2)$$

Here μ_{original} is the exponential growth of the *uncorrected equality recurrence*, not the polyomino growth constant λ . In particular, the lower inequality in (2) is *not* a lower bound on λ .

The third result describes both the power and a limitation of exact-prefix corrections. The corrected majorants decrease coefficientwise as more exact terms are supplied. They eventually admit certificates at any evaluation point where the linearized system at the true series has spectral radius less than one. Conversely, a supercritical linearization obstructs the whole hierarchy under a mild nonzero-tail condition. A scalar example has true growth 1 but corrected-majorant growth tending to 3. [Added 29 September 2026, batch 40: Part II classifies the equality case for an irreducible true-profile Jacobian, identifies the limiting radius of the hierarchy, and proves that for the fixed Bui map every corrected majorant has growth at least 4.3149 (theorems 15.2, 16.3 and 22.2). That number is a limit of the method, not a lower bound on λ .]

1.3 Novelty and dependency boundaries

The proposed research contributions are the explicit 4.498 polyomino bound, the finite-prefix application and its exact data, the arithmetic response to Bui’s certificate question, and the accompanying analysis of the prefix hierarchy. The examined repository snapshot and primary sources did not contain this 4.498 certificate or this marked-prefix calculation. That is a limited novelty assessment, not an exhaustive priority determination.

Several ingredients are established mathematics. The geometric inequalities come from Bui. Enumeration by canonical include/exclude growth is in the tradition of Redelmeier [11]. Positive monomial inequalities, logarithmic convexification, and entropy-like dual expressions belong to geometric programming [13, 14, 15]. We do not present those tools as new discoveries. The proof of the certificate alternative is included to make its hypotheses, strictness, and rational verification completely explicit in the recurrence setting.

Three distinct levels. The geometric recurrence theorem is an imported mathematical input. The new prefix and dual-certificate theorems have written proofs here. The headline numerical bound additionally uses the finite enumeration in Appendix A, whose source programs and exact outputs are supplied. Arithmetic verification is not a substitute for verifying enumeration.

[write] In ProveIt the first level is stronger than “imported”: it is a kernel-checked Lean theorem about the actual counts (section 1.5). The second and third levels are not formalized.

1.4 [write] Notation conventions

This subsection was added when the manuscript was written into ProveIt. No delivered symbol was renamed and no normalization was changed; the conventions below only fix how the delivered symbols are to be read.

Coordinate letters. The seventeen upper-case letters of (4) name the marked neighborhood types and their counting sequences ($C_n, \dots, Z_n$, also written $a_{i,n}$ with $i \in \mathcal{I}$); the matching lower-case letters $c, \dots, z$ are the coordinates of the vector X in (6) and the row names of table 4. A coordinate is always a bare letter or a subscript index ($v_G, a_{D,7}, \Phi_p$). Many of these letters are reused for other objects, which are recognized by a subscript N , by calligraphic type, or by the local context listed here:

Letter	Other meanings in this report
P	$P_N = [\xi^{\leq N}]a$, the exact prefix; a polyomino P (section 2 and proposition 4.2)
D	D_N , the finite defect; the defect polynomial D in (24)
F	$F(X) = \Phi(\zeta, X)$ with components F_i , and $f_i(y) = \log F_i(e^y) - y_i$ in (19) (section 6)
X, d	the unknowns $X = (X_1, \dots, X_d)$ of a general system $\Phi \in \mathbb{R}_{\geq 0}[\xi, X_1, \dots, X_d]^d$ (sections 3, 5 and 6), e.g. $X \in \mathbb{R}_{>0}^d$ in the sets S_K of lemma 6.2; the bold X is the vector of (6)
T, Y	$T_N = [\xi^{\leq N}]\Phi(\xi, P_N)$; $Y_N = B_N - P_N$
E	forbidden offset set; the enumerator's excluded set; the defect improvement in (24)
R, S	required offset set, radii R_i ; the enumerator's occupied set, the sets S_K of lemma 6.2
U	the enumerator's frontier; $U = a - P_N$ and a generic subsolution (lemma 3.2 and theorem 3.3)
H	the nonlinear part in $\Phi = LX + H$; the half-plane $\mathcal{H}$
B	the offset sets B, B_4 ; the majorants B_N ; a least fixed point B (section 5)
C	the dual product C ; a constant C (theorem 8.1); the normalizer $C = \sum_i \ell_i \tau_i$
h, q	$h(y)$ of (19), $h = (I - J)^{-1}\mathbf{1}$, a row index; the argument of $\log q$, a degree $q \geq 2$
$v, w, y, p, t, u, x, e,$	certificate vector $v, w = v - P_N(\zeta)$, log-coordinates $y, p_N = P_N(\zeta)$, reals $t, u,$
c, g, r	scalar unknown x , unit vectors e_i , coefficients c_{ik} , a nearest point g , row sums r_i
b	b_n (proof of lemma 3.2); exponents b_{ik} in (15); forcing vector b_N (section 8)

Collisions with the ProveIt project. The project's research note [7] and its Lean files use the same λ , the same $A(n) = A_n$, the same seventeen upper-case sequence names with lower-case coordinates, and the same evaluation point ζ ; its numeric map Φ_ζ is $\Phi(\zeta, \cdot)$ here. Two symbols differ:

- The project writes $b_N = (c_N, \dots, z_N)$ for the ζ -weighted prefix profile and $\mathcal{S}_N = \sum_{n \leq N} S(n)\zeta^n$ for one sequence. That profile is $p_N = P_N(\zeta)$ here. *Tempting false reading:* the b_N of section 8 is the project's b_N . It is not; it is the high-degree forcing vector $([\xi^{>N}]\Phi(\xi, P_N))(\zeta)$.
- The project writes a for its rational supersolution; that is v here, while $a = a(\zeta)$ in section 8.

The symbol μ (the growth of an equality recurrence, μ_{original} for Bui's uncorrected system) occurs only in this report. *Tempting false reading:* $\mu_{\text{original}} \geq 4.52349$ is a lower bound on λ . It is not (section 5). The convention $A_0 = 0$ of section 2 differs from the value $a(0) = 1$ of OEIS A001168; positive-size terms agree.

1.5 [write] What ProveIt has formalized, and what it has not

This report belongs to ProveIt's research-report collection and continues the formal project Combinatorics/Polyominoes/KlarnerConstant/; being filed next to, or citing, a Lean development confers no formal status on it. That project proves the following in Lean (namespace LeanProofs.KlarnerConstant; files under Combinatorics/Polyominoes/KlarnerConstant/Lean/KlarnerConstant/, cited below by file name), with no sorry, project axiom or native_decide; its assumption audit is Audit.lean. Line numbers are those of the pin; the project's files are unchanged between the pin and the placement commit.

- *Bui's Lemma 4 for the actual counts.* geometricPublishedBuiRecurrences (GeometricComplete.lean:63) proves PublishedBuiRecurrences for geometricCoefficientProfile: the seventeen recurrences for $n \geq 2$, the published degree-one values, nonnegativity and vanishing at zero, for the finite marked-occurrence counts of the patterns in Patterns.lean. Coefficient by coefficient this is the inequality (7), which the delivered text calls imported; reading the Lean structure as (7) in formal-series notation is not itself a formal step.

- *The five partitions (8)* as equalities: `buiF_occurrenceCount_eq_g_add_p`, `buiG_occurrenceCount_eq_e_add_q`, `buiH_occurrenceCount_eq_d_add_s`, `buiR_occurrenceCount_eq_y_add_w` and `buiT_occurrenceCount_eq_x_add_v` (`GeometricLinear.lean:279-312`).
- *G-domination* $A_n \leq G_n$: `fixedPolyominoCount_le_geometricCoefficientProfile_g` (`GeometricProfile.lean:371`). The other half $G_n \leq nA_n$ of (5) is not formalized; the upper bound does not need it.
- *Supermultiplicativity and Fekete (3)*: `fixedPolyominoCount_supermultiplicative` and `tendsto_realNthRoot_fixedPolyominoCount_growthSup` (`GeometricComplete.lean:97`).
- *The Lean-verified bound* $\lambda \leq 9047/2000 = 4.5235$: `certificate_isSupersolution` (`Certificate.lean:232`), `fixedPolyominoCount_le_9047_div_2000_pow` (`GeometricComplete.lean:69`), its real cast (`:76`) and `growthSup_fixedPolyominoCount_le_9047_div_2000` (`:88`).

The one human boundary of that development is the transcription of Bui’s page-15 diagrams into the offset tables of `Patterns.lean`.

Nothing new in this report is formalized: not [lemma 3.2](#), [theorems 3.3](#), [3.4](#) and [3.6](#), not the size- ≤ 18 table, not [theorem 4.1](#) ($\lambda \leq 2249/500 = 4.498$, which is computer-assisted), and none of [sections 5](#) to [8](#). [Added 29 September 2026, batch 40: nor anything in Part II, including the 4.3149 barrier of [theorem 22.2](#).] The project’s endpoint theorems cannot simply be re-instantiated: they are hard-wired to $\zeta = 2000/9047$ (`certificateZeta`) in `PrefixRecurrence.g_le_certificate` and `g_lt_one` (`Recurrence.lean:321, 327`), `CoefficientProfile.g_lt_9047_div_2000_pow` (`:457`) and `dominatedCoefficient_le_9047_div_2000_pow` (`:477`). Only `buiIterate_le_supersolution` (`:283`) and `PrefixRecurrence.le_supersolution` (`:309`) are generic in ζ . A formal 4.498 bound therefore needs, beyond the plan of [section 9](#), a prefix-corrected variant of that weighted-prefix engine and a computable bridge to the noncomputable `BuiNeighborhood.occurrenceCount` (`Patterns.lean:140`). As a formalization target, not a claim: the certificates for $N = 7, \dots, 10$ in [table 3](#) (4.5233 down to 4.5194) already improve on 4.5235 and need exact marked counts only through size 7 to 10.

The trust boundary of [theorem 4.1](#) is thus: Lean-proved recurrences (relative to `Patterns.lean`), plus an exhaustive enumeration that is not kernel-checked (regenerated at intake, [section D](#)), plus exactly replayed rational arithmetic (Python, not Lean), plus the written and unrefereed proofs of [section 3](#).

2 Polyomino counts and the positive polynomial system

2.1 Counting conventions

A polyomino is a nonempty finite edge-connected subset of $\mathbb{Z}^2$. Two such subsets are equivalent precisely when one is a translate of the other. Rotations and reflections are not identified. Holes are allowed. Let A_n be the number of equivalence classes with n cells, for $n \geq 1$. For the recurrence calculation we put $A_0 = 0$ and all marked size-zero counts equal to zero. This convention differs from the optional empty-object term $A_0 = 1$ in some sequence tables; positive-size terms are unchanged.

The standard concatenation argument gives $A_{m+n} \geq A_m A_n$ for positive m, n . Fekete’s lemma then gives

$$\lambda = \lim_{n \rightarrow \infty} A_n^{1/n} = \sup_{n \geq 1} A_n^{1/n}. \quad (3)$$

These are background inputs also implemented in the repository’s `Concatenation.lean` and

Asymptotic.lean [1, 3, 6].

A neighborhood pattern consists of a required offset set R containing $(0, 0)$ and a disjoint forbidden offset set E . An occurrence at a cell a of a polyomino P means

$$a + R \subseteq P, \quad (a + E) \cap P = \emptyset.$$

The marked count for a pattern is the sum of its occurrences over all fixed polyominoes of the given size. Counting occurrences, rather than merely polyominoes which have an occurrence, is essential.

Write $C_n, D_n, \dots, Z_n$ for the seventeen counts, in the coordinate order

$$I = (C, D, E, F, G, H, P, Q, R, S, T, U, V, W, X, Y, Z). \quad (4)$$

To avoid confusing a coordinate name with a function, the polynomial map will be denoted by Φ .

2.2 Exact offset specification

Let

$$B = \{(-1, -1), (0, -1), (1, -1)\}, \quad B_4 = B \cup \{(2, -1)\}.$$

In table 1, 1, 2, 3 in the required column mean respectively $\{(0, 0)\}$, $\{(0, 0), (1, 0)\}$, and $\{(0, 0), (1, 0), (2, 0)\}$. A blank lattice position imposes no condition. These coordinate sets agree with the repository's pattern table and Bui's page-15 neighborhood diagrams [5, 1].

Type	Required	Forbidden offsets
C	1	$B \cup \{(-1, 1), (1, 1), (-1, 0), (1, 0)\}$
D	1	$B \cup \{(-1, 1), (-1, 0), (1, 0)\}$
E	1	$B \cup \{(-1, 0), (1, 0)\}$
F	1	B
G	1	$B \cup \{(-1, 0)\}$
H	1	$B \cup \{(-1, 1), (-1, 0)\}$
P	2	$\{(0, -1), (1, -1), (2, -1)\}$
Q	2	$B \cup \{(-1, 0)\}$
R	2	$B_4 \cup \{(-1, 1), (-1, 0)\}$
S	2	$B \cup \{(-1, 1), (-1, 0)\}$
T	2	$B_4 \cup \{(-1, 0)\}$
U	2	B_4
V	3	$B_4 \cup \{(-1, 0)\}$
W	3	$B_4 \cup \{(-1, 1), (-1, 0)\}$
X	2	$B_4 \cup \{(-1, 0), (2, 0)\}$
Y	2	$B_4 \cup \{(-1, 1), (-1, 0), (2, 0)\}$
Z	2	$B_4 \cup \{(-1, 1), (2, 1), (-1, 0), (2, 0)\}$

Table 1: The seventeen marked neighborhood patterns.

Every polyomino has a G occurrence at the leftmost cell of its bottom row. There are at most n possible marked cells. Consequently

$$A_n \leq G_n \leq nA_n. \quad (5)$$

The first six size-one counts equal one; the other eleven equal zero.

2.3 The imported coefficient inequalities

Use the formal variable ξ and coordinate vector $\mathbf{X} = (c, d, e, f, g, h, p, q, r, s, t, u, v, w, x, y, z)$. Define

$$\begin{aligned}
 \Phi_C &= \xi + \xi e, & \Phi_D &= \xi + \xi g, & \Phi_E &= \xi + \xi f, \\
 \Phi_F &= g + p, & \Phi_G &= e + q, & \Phi_H &= d + s, \\
 \Phi_P &= eh + qd + xr + vy + uyz, \\
 \Phi_Q &= \xi g + \xi ge + \xi^2 u + \xi^2 t g + \xi^2 ru, \\
 \Phi_R &= y + w, \\
 \Phi_S &= \xi g + \xi e^2 + \xi^2 t + \xi^2 x g + \xi^2 y u, \\
 \Phi_T &= x + v, \\
 \Phi_U &= dh + sd + yr + wy + uz^2, \\
 \Phi_V &= \xi s + \xi^2 g^2 + \xi^2 te + \xi^2 rt, \\
 \Phi_W &= \xi s + \xi^2 e g + \xi^2 x e + \xi^2 y t, \\
 \Phi_X &= \xi d + \xi^2 g + \xi^2 u, \\
 \Phi_Y &= \xi c + \xi^2 g + \xi^2 t, \\
 \Phi_Z &= \xi c + \xi^2 e + \xi^2 x.
 \end{aligned} \tag{6}$$

There are 53 monomials, each with coefficient one. The order displayed within each row is also the order of the dual weights in [table 4](#). In the first two displayed lines, each coordinate has its own two-term row.

For the true generating series $\mathbf{a}_i(\xi) = \sum_{n \geq 1} a_{i,n} \xi^n$, Bui’s geometric lemma is the coefficientwise inequality

$$\mathbf{a} \leq \Phi(\xi, \mathbf{a}). \tag{7}$$

All products in this equation are formal Cauchy products. Thus convergence is not part of the statement. Equations with a factor ξ^2 , for example, encode a two-cell shift in the corresponding convolution index.

The geometric proof of (7) uses occupancy partitions, injective leaf deletions, and decompositions into connected marked pieces. We import that lemma rather than silently treating a numerical recurrence fit as a geometric proof. For example, the C row deletes its marked cell, whose only possible neighbor is above it. The P and U rows encode multi-piece decompositions; these are not inferred from the finite table. The repository supplies their separate finite-injection modules [1, 6]. [write] Those modules prove the lemma in Lean for the actual counts (`geometricPublishedBuiRecurrences`); so, within ProveIt, the input (7) is a kernel-checked theorem relative to the offset tables of `Patterns.lean`, not an imported assumption ([section 1.5](#)).

The five same-size partitions are in fact exact:

$$F = G + P, \quad G = E + Q, \quad H = D + S, \quad R = Y + W, \quad T = X + V. \tag{8}$$

For example, a type- G anchor either has no right neighbor, yielding type E , or has a right neighbor, yielding type Q . Similar left/right occupancy splits give the other identities. They will serve as independent consistency checks on the occurrence table. [write] The five identities are Lean theorems in ProveIt (`GeometricLinear.lean:279–312`); the table’s agreement with them is checked by the Python verifier, not by Lean.

3 Exact-prefix correction: the general theorem

3.1 Proper positive polynomial systems

Definition 3.1. A proper positive polynomial system is a map $\Phi \in \mathbb{R}_{\geq 0}[\xi, X_1, \dots, X_d]^d$ such that $\Phi(0, 0) = 0$ and the nonnegative matrix

$$L = \Phi_X(0, 0)$$

is nilpotent. Here and below vector inequalities are coordinatewise.

Nilpotence handles same-size linear dependencies. In (6), the only such dependencies are the five rows in (8); their directed graph has no cycle. All remaining occurrences of an unknown are multiplied either by a positive power of ξ or by another unknown. Thus the system is proper. One possible degree-by-degree evaluation order is

$$C, D, E, Q, S, V, W, X, Y, Z, P, U, G, F, H, T, R.$$

The formal development does not require this particular order.

Lemma 3.2 (Formal fixed points and comparison). *A proper positive polynomial system has a unique fixed point in $\xi\mathbb{R}[[\xi]]^d$. Its coefficients are nonnegative, and the iterates from zero increase coefficientwise and stabilize at each finite degree. If $U \in \xi\mathbb{R}_{\geq 0}[[\xi]]^d$ satisfies $U \leq \Phi(\xi, U)$, then U is coefficientwise at most this fixed point.*

Proof. Write $\Phi = LX + H$. Because the unknown series have zero constant terms, the coefficient of degree n in $H(\xi, X)$ depends only on unknown coefficients of degrees strictly less than n . If these have already been chosen, the coefficient vector b_n in a fixed point must satisfy

$$(I - L)b_n = h_n.$$

For some k , $L^k = 0$, so

$$(I - L)^{-1} = I + L + \dots + L^{k-1} \geq 0.$$

This proves existence, uniqueness, and nonnegativity by induction on n . For a subsolution, the same argument gives $(I - L)u_n \leq h_n(u_1, \dots, u_{n-1})$; multiplication by the nonnegative inverse and induction prove comparison.

The iterates from zero are increasing by monotonicity. If all lower-degree coefficients have stabilized, the remaining degree- n error is multiplied by L at each further iteration, and therefore vanishes after at most k steps. This proves finite-degree stabilization. $\square$

The subsolution conclusion uses formal properness. It would be false for arbitrary numerical subsolutions: $2 \leq 2^2$ does not imply that 2 is below the least nonnegative fixed point of $x \mapsto x^2$.

3.2 The finite defect and the nonnegative tail map

Suppose $a \in \xi\mathbb{R}_{\geq 0}[[\xi]]^d$ is a true counting series satisfying (7). For $N \geq 1$, set

$$P_N = [\xi^{\leq N}]a, \quad T_N = [\xi^{\leq N}]\Phi(\xi, P_N), \quad D_N = T_N - P_N. \quad (9)$$

The coefficient of $\Phi(\xi, a)$ at a degree at most N depends only on P_N , so $D_N \geq 0$. It is precisely the finite prefix of the true recurrence defect $\Phi(\xi, a) - a$.

The signed equation

$$B_N = \Phi(\xi, B_N) - D_N \quad (10)$$

is a convenient shorthand, but its right-hand side is not a positive polynomial in ξ . The correct positivity argument uses $Y_N = B_N - P_N$ and the map

$$\begin{aligned}\Psi_N(\xi, Y) &= \Phi(\xi, P_N + Y) - T_N \\ &= \Phi(\xi, P_N + Y) - \Phi(\xi, P_N) + [\xi^{>N}] \Phi(\xi, P_N).\end{aligned}\tag{11}$$

Every coefficient of Ψ_N in $\xi, Y_1, \dots, Y_d$ is nonnegative. Indeed, in the difference on the second line, the terms independent of Y cancel, leaving only monomials containing at least one Y . Moreover, $\Psi_N(0, 0) = 0$ and $(\Psi_N)_Y(0, 0) = L$.

Theorem 3.3 (Prefix-corrected majorant). *The map Ψ_N has a unique fixed point $Y_N \in \xi \mathbb{R}_{\geq 0}[[\xi]]^d$. Its coefficients through degree N vanish. The series $B_N = P_N + Y_N$ satisfies*

$$a \leq B_N, \quad [\xi^{\leq N}] B_N = P_N,$$

and solves (10).

Proof. The fixed-point statement follows from lemma 3.2. The constant-in- Y part of Ψ_N begins above degree N , so degree induction gives $[\xi^n] Y_N = 0$ for $n \leq N$. Put $U = a - P_N \geq 0$. The low-degree part of $\Phi(\xi, a) - a$ is D_N , and its remaining coefficients are nonnegative. Therefore

$$U \leq \Phi(\xi, P_N + U) - T_N = \Psi_N(\xi, U).$$

Formal comparison yields $U \leq Y_N$. Adding P_N proves domination. Substituting the fixed-point equation for Y_N yields (10). $\square$

Theorem 3.4 (Rational upper certificate). *Let $\zeta > 0$. Suppose a vector $v \in \mathbb{R}^d$ satisfies*

$$v \geq P_N(\zeta), \quad v \geq \Phi(\zeta, v) - D_N(\zeta).\tag{12}$$

Then all coordinates of B_N converge at ζ , and

$$B_N(\zeta) \leq v, \quad a_{i,n} \leq v_i \zeta^{-n}.$$

For the polyomino system this implies $A_n \leq v_G \zeta^{-n}$ and $\lambda \leq \zeta^{-1}$. When the input prefix, ζ , and v are rational, verification of (12) requires rational arithmetic only.

Proof. Let $w = v - P_N(\zeta) \geq 0$. Equations (9) and (11) show that (12) is equivalent to

$$\Psi_N(\zeta, w) \leq w.$$

The numerical iterates $w^{(0)} = 0$, $w^{(j+1)} = \Psi_N(\zeta, w^{(j)})$ therefore stay between 0 and w . The corresponding formal iterates are polynomials with nonnegative coefficients. Their evaluation at ζ is exactly $w^{(j)}$. For every fixed degree M , a sufficiently late formal iterate agrees with Y_N through degree M by lemma 3.2. Hence

$$\sum_{n=0}^M [\xi^n] Y_N \zeta^n \leq w.$$

Monotone convergence of these bounded finite sums proves $Y_N(\zeta) \leq w$. Adding $P_N(\zeta)$ proves the asserted bound for B_N . Every nonnegative term of a bounded series is bounded by its sum, yielding the coefficient estimate. Now use (5) and (3). $\square$

Remark 3.5 (The tail condition is essential). A solution of $v \geq \Phi(\zeta, v) - D_N(\zeta)$ alone need not be a valid certificate. The subtraction can create extraneous branches or vectors below the known prefix. The separate inequality $v \geq P_N(\zeta)$ is part of the proof and is checked explicitly by the verifier.

3.3 Monotonicity of the hierarchy

Theorem 3.6 (More exact data never worsens the majorant). *For every $N \geq 1$,*

$$a \leq B_{N+1} \leq B_N.$$

Thus the radii of convergence of the corrected majorants are nondecreasing, and their reciprocal exponential-growth bounds are nonincreasing. Any real upper certificate for prefix N is also an upper certificate for prefix $N + 1$ at the same evaluation point.

Proof. The first inequality is [theorem 3.3](#). Through degree N , the two majorants agree. At degree $N + 1$, B_{N+1} has the true coefficient and B_N has a coefficient at least as large. Above that degree, both obey the same positive coefficient recurrence, because the defect polynomials have no terms there. Degree induction using $(I - L)^{-1} \geq 0$ proves the second inequality. Coefficientwise domination implies the radius assertion.

If v is an N -certificate, then the preceding theorem and $a \leq B_N$ show that $P_{N+1}(\zeta) \leq v$. Also $D_{N+1} - D_N$ has nonnegative coefficients, so $\Phi(\zeta, v) - D_{N+1}(\zeta) \leq v$. Both certificate conditions for $N + 1$ follow. $\square$

For the present system the defect is zero through size 6. Its size-7 coefficient, in the order (4), is

$$[\xi^7]D_7 = (0, 1, 2, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 2). \quad (13)$$

For example, $a_{D,7} = 260$, whereas the deletion bound uses $a_{G,6} = 261$; the corresponding defect coefficient is 1. This first discrepancy explains why prefix lengths 1 through 6 do not improve the original equality system.

4 An exact upper bound below 4.5

4.1 The certificate

Take $N = 18$ and

$$\zeta = \frac{500}{2249}.$$

Use the exact marked counts in [Appendix A](#) to construct P_{18}, T_{18}, D_{18} by (9). The vector v is given in [table 2](#); every numerator in its second column has common denominator 10^9 .

Define the residual and tail budget by

$$\sigma_i = v_i - \Phi_i(\zeta, v) + D_{18,i}(\zeta), \quad \eta_i = v_i - P_{18,i}(\zeta). \quad (14)$$

The last two columns are conservative lower bounds:

$$\sigma_i \geq \frac{\text{third-column entry}}{10^{12}}, \quad \eta_i \geq \frac{\text{fourth-column entry}}{10^6}.$$

They are obtained by flooring the exact rational numbers after scaling; no printed decimal approximation is being used to decide a sign.

Theorem 4.1 (Computer-assisted polyomino bound). *For the fixed-polyomino counting convention in [section 2](#),*

$$A_n \leq \frac{927884613}{10^9} \left(\frac{2249}{500} \right)^n \quad (n \geq 1), \quad \lambda \leq 4.498 < 4.5.$$

Coordinate	Numerator of v_i	10^{12} -scaled slack	10^6 -scaled tail
c	347,752,792	99,824	28,076
d	427,615,666	100,303	59,180
e	564,832,205	100,345	123,441
f	1,551,872,652	100,000	542,918
g	927,884,613	99,000	260,176
h	727,305,121	100,000	159,091
p	623,987,939	99,749	282,741
q	363,052,309	100,459	136,735
r	234,092,913	100,000	63,133
s	299,689,355	99,657	99,910
t	283,350,363	100,000	89,179
u	487,086,219	100,056	192,950
v	119,025,613	99,875	52,209
w	97,883,000	99,767	38,611
x	164,324,650	100,195	36,970
y	136,209,813	99,861	24,521
z	112,439,657	100,134	14,724

Table 2: The rational size-18 upper certificate and exact lower bounds on its margins. All vector entries have denominator 10^9 .

Proof. The finite enumeration described below produces the true coefficients of P_{18} in Appendix A. Bui’s recurrence lemma gives (7). Exact integer polynomial convolution constructs D_{18} . Substituting the rational vector of table 2 into (14) gives

$$\min_i \sigma_i \geq \frac{99}{10^9} > 0, \quad \min_i \eta_i \geq \frac{14724}{10^6} > 0.$$

Thus theorem 3.4 applies. Its G coordinate is $v_G = 927884613/10^9$, yielding the claimed inequalities. $\square$

The strict residuals also show that the certificate is not balanced on a rounding boundary. By continuity, the same vector works for a sufficiently small increase of ζ . We retain 2249/500 as the stated bound because it is simple and explicitly verified.

4.2 Discovery is not verification

The search for a candidate used floating-point solutions of

$$\Phi(\zeta, v) - D_N(\zeta) = v, \quad \det(I - \Phi_X(\zeta, v)) = 0.$$

These equations suggest where the equality envelope becomes critical. For a slightly smaller evaluation point, we solved a perturbed fixed-point equation with a small positive residual in every coordinate, then rounded the candidate to denominator 10^9 . Such a solver may converge to an undesired branch or misestimate a threshold. Neither event can invalidate an accepted certificate: the verifier checks the two inequalities in (12) independently from scratch.

Table 3 records the observed progression. The middle column contains numerical critical-point estimates, not proved optimal values or certified lower bounds. Every entry in the last column has a separate exact rational certificate in the archive.

The original recurrence’s growth is already within 10^{-5} of the repository’s bound, as section 7 will prove. Consequently, theorem 4.1 does not arise from finding a better supersolution for the unchanged system. The correction changes the equality envelope while preserving its validity as an upper bound on the true counts.

Prefix size N	Estimated equality growth	Certified upper bound
1–6	4.523499228	4.5235 (original certificate)
7	4.523230947	4.5233
8	4.522434056	4.5225
9	4.521121227	4.5212
10	4.519368978	4.5194
11	4.517262575	4.5173
12	4.514881253	4.5149
13	4.512294495	4.5123
14	4.509560290	4.5096
15	4.506725656	4.5068
16	4.503828169	4.5039
17	4.500897596	4.5009
18	4.497957351	4.4980

Table 3: Progress under exact-prefix correction. Numerical estimates and rigorous upper certificates are deliberately separated.

4.3 A reproducible finite enumeration

Every translation class has a unique representative whose leftmost cell on the bottom row is $(0, 0)$. All its cells then lie in

$$\mathcal{H} = \{(x, y) : y > 0\} \cup \{(x, 0) : x \geq 0\}.$$

The main enumerator maintains a connected occupied set S , a finite ordered frontier U of available adjacent cells, and an excluded set E . Initially $S = \emptyset$, $U = \{(0, 0)\}$, and $E = \emptyset$. At a call, take the last frontier cell p , include it in S , record the resulting object, and recurse after adding newly available neighbors. After that branch returns, remove p and exclude it for all later branches of the same call. The exclusions introduced by a call are removed when that call returns. No quotient by rotations or reflections is performed.

Proposition 4.2 (Enumeration correctness). *The include/exclude frontier procedure records each nonempty finite edge-connected subset of $\mathcal{H}$ containing $(0, 0)$ exactly once, up to the specified size limit. Thus its totals are precisely the fixed-polyomino counts.*

Proof. Every recorded set is connected because each added cell is adjacent to the previous occupied set, except for the initial origin. It stays in $\mathcal{H}$ by the admissibility test.

Fix a target set P of the required kind. At any intermediate state with $S \subsetneq P$ and no excluded cell in P , connectedness implies that some cell of $P \setminus S$ lies in the available frontier. In the order processed by the current call, there is a unique first frontier cell belonging to P . All cells processed before it can be excluded without affecting P . The branch including that unique cell preserves the same invariant for P . Induction reaches $S = P$, which is recorded.

The same argument proves uniqueness: the first processed frontier cell belonging to P determines the only branch that can contain P without excluding one of its cells. Applying this observation recursively determines a unique path. Finally, the chosen anchor gives exactly one representative of each translation class. $\square$

For an occupied anchor, its membership in each pattern depends only on ten nearby positions:

$$(-1, 1), (1, 1), (2, 1), (-1, 0), (1, 0), (2, 0), (-1, -1), (0, -1), (1, -1), (2, -1).$$

These are encoded by a ten-bit mask. All 2^{10} masks are preclassified against the required and forbidden sets of [table 1](#). The direct implementation scans all occupied anchors whenever an

object is recorded. The faster implementation maintains the pattern totals under cell insertion and deletion. Inserting p changes only the contribution of p and those anchors $p - o$ for which o is one of the ten offsets. Removing p reverses exactly these changes. This local invariant proves that incremental updates give the same occurrence count as a complete scan.

The size-18 computation records

$$A_{18} = 1,540,820,542, \quad \sum_{n=1}^{18} A_n = 2,083,404,030$$

objects across all recorded sizes. The unmarked totals through 18 agree with OEIS A001168 [12]; those familiar totals are a consistency check, not a replacement for the marked enumeration.

4.4 Finite-machine safety and independent checks

The C++ sources are restricted to size at most 18. Any cell of an anchored n -cell polyomino satisfies $|x| \leq n - 1$ and $0 \leq y \leq n - 1$. The padded array is wide and tall enough for these coordinates and for all mask and neighbor offsets. Thus no boundary wrapping or out-of-range access is needed by the algorithm.

The counters are unsigned 64-bit integers. An a priori bound verifies that overflow is impossible at these sizes. A deterministic depth-first spanning tree of an anchored n -cell set can be encoded by a plane rooted tree and directions for its $n - 1$ forward steps. There are Cat_{n-1} plane tree shapes. The first forward step has at most four choices; every later forward step has at most three, since a nonroot vertex cannot move back to its parent and the root has already used one child direction. Consequently, for $n \geq 2$,

$$A_n \leq 4 \cdot 3^{n-2} \text{Cat}_{n-1}.$$

This overcounts heavily but is sufficient. Every marked count is at most nA_n , and the largest bound for $n \leq 18$ is

$$18 \cdot 4 \cdot 3^{16} \text{Cat}_{17} = 401816383504818480 < 2^{64}.$$

Incremental counters count occurrences in the currently occupied finite set; the insertion/deletion invariant also prevents underflow.

Two structurally different enumeration approaches were compared through size 11: the C++ include/exclude procedure and a Python procedure that repeatedly adds each boundary cell, translates each shape to a canonical coordinate tuple, and deduplicates using a set. All 187 marked values and all unmarked totals agreed. Direct scanning and incremental scanning in C++ agreed through size 16. The final size-18 output also agrees with the previous size-16 table on every earlier entry.

These checks do not amount to independent formal certification of all size-18 values. They locate the remaining computational trust boundary: the correctness of the supplied enumerator, its implementation, and its execution. The recurrence inequalities themselves are not learned from this table. Five exact partition identities and all nonnegative recurrence defects are checked again by the rational verifier.

[write] *Regeneration at intake*. The delivered validation record says that no fresh size-18 run was made after the enumerators' final command-line changes. When the manuscript was received by ProveIt, the delivered code/enumerate.cpp was compiled afresh (c++ -O3 -std=c++17, g++ 16.1.0) and run as `enumerate 18` (8 min 23 s on one core of the intake machine); after CSV parsing its output equals the shipped data/profiles_18.csv in all 19 columns of all 18 rows (the files differ only in line endings). Fresh runs through sizes 11–14, 16 and 17 matched as well, and an independent set-based Python enumerator, with pattern coordinates taken from the project's `Patterns.lean` rather than from this package, matched all 10 unmarked and 170 marked entries

through size 10. So the size-18 table is now reproduced from the final source. This does not change the boundary stated above: the regeneration uses the same program and algorithm, the rational verifier checks arithmetic and necessary consistency conditions only, and the correctness of the enumeration rests on [proposition 4.2](#) and the incremental-update argument together with this regeneration.

5 Balanced-monomial certificates of divergence

5.1 The certificate format

Consider a finite positive polynomial system

$$\Phi_i(\zeta, X) = \sum_{k \in K_i} c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}}, \quad c_{ik} \in \mathbb{Q}_{>0}, \quad b_{ik} \in \mathbb{N}, \quad \alpha_{ik} \in \mathbb{N}^d. \quad (15)$$

Fix a rational $\zeta > 0$. We wish to certify that there is no $X \in \mathbb{R}_{>0}^d$ satisfying $\Phi(\zeta, X) \leq X$.

Assign a nonnegative integer weight m_{ik} to each monomial, not all zero, and define row sums

$$r_i = \sum_{k \in K_i} m_{ik}.$$

The weights are *balanced* if

$$\sum_{i,k} m_{ik} \alpha_{ik,j} = r_j \quad (1 \leq j \leq d). \quad (16)$$

Define the positive rational product

$$C(\zeta, m) = \prod_{i,k:m_{ik}>0} \left(\frac{c_{ik} \zeta^{b_{ik}} r_i}{m_{ik}} \right)^{m_{ik}}. \quad (17)$$

A row of weight zero contributes no factor. The entire certificate consists of ζ , the integer weights, and a verification that $C(\zeta, m) > 1$.

Theorem 5.1 (Balanced-monomial obstruction). *If (16) holds and $C(\zeta, m) > 1$, then no positive supersolution of $\Phi(\zeta, X) \leq X$ exists.*

Proof. Suppose such a supersolution exists. For each row with $r_i > 0$, put $p_{ik} = m_{ik}/r_i$ on the positive-weight terms. Weighted arithmetic–geometric mean gives

$$\begin{aligned} 1 &\geq \frac{\Phi_i(\zeta, X)}{X_i} \geq \sum_{k:m_{ik}>0} c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}-e_i} \\ &\geq \prod_{k:m_{ik}>0} \left(\frac{c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}-e_i}}{p_{ik}} \right)^{p_{ik}}. \end{aligned}$$

Raise the inequality to r_i and multiply over rows. The exponent of X_j in the resulting right-hand side is $\sum_{i,k} m_{ik} \alpha_{ik,j} - r_j = 0$ by balance. All unknown variables cancel, leaving $1 \geq C(\zeta, m) > 1$, a contradiction. $\square$

The cancellation is the essential point. A certificate does not need to supply a very large iterate, an approximate divergent orbit, or an approximate critical point. Its inequalities quantify over every positive candidate supersolution at once.

Example 5.2 (Catalan recurrence). For $\Phi(\xi, x) = \xi + x^2$, assign weight one to both terms. Then the incoming exponent is $0 + 2 = 2 = r$, and

$$C(\zeta, m) = (2\zeta)(2) = 4\zeta.$$

Thus $\zeta > 1/4$ immediately has a two-weight divergence certificate, recovering the usual Catalan threshold from one arithmetic–geometric mean inequality.

5.2 From nonexistence to a lower growth bound

Let B be the least formal fixed point of a proper positive system. If its coordinates converge at ζ , finite products of nonnegative convergent series can be evaluated, giving

$$B(\zeta) = \Phi(\zeta, B(\zeta)).$$

If the system is productive in the sense defined in the next section, this fixed point is strictly positive. Therefore [theorem 5.1](#) implies that at least one coordinate fails to converge at ζ .

Define the system’s exponential growth by

$$\mu = \max_i \limsup_{n \rightarrow \infty} ([\xi^n] B_i)^{1/n}.$$

Equivalently, $1/\mu$ is the minimum of the coordinate radii of convergence. Cauchy–Hadamard then gives

$$\mu \geq \zeta^{-1}. \tag{18}$$

We use the limsup formulation so that no unproved assertion about periodicity or existence of individual root limits is needed.

For productive systems with a strongly connected dependency graph, the coordinate radii coincide. Indeed, for every dependency edge $i \rightarrow j$, choose one positive coefficient in each of the other factors of a monomial containing X_j . The formal fixed-point equation implies $B_i \geq c \xi^k B_j$ for some $c > 0$ and integer $k \geq 0$. Thus $R_i \leq R_j$. Following paths in both directions proves equality.

Do not reverse the majorant inequality. The true polyomino series is below the equality solution, so an upper bound on equality growth bounds λ from above. A lower bound on equality growth does not bound λ from below. The dual certificate is useful for certifying the limitation of a chosen recurrence model and for answering the recurrence-growth question, not for reversing this implication.

6 Completeness of the certificate alternative

6.1 The hypotheses

For this section fix $\zeta > 0$ and write $F(X) = \Phi(\zeta, X)$. The coefficients of F are positive on its listed monomials.

Definition 6.1. The map F is *productive* if some finite iterate of F , starting from zero, has every coordinate strictly positive. Its dependency graph has an edge $i \rightarrow j$ if some monomial of F_i contains X_j . The map is *strongly connected* if this graph is strongly connected, and *nonlinear* if at least one monomial has total degree at least two in the unknowns.

Productivity is stronger than merely having a constant term somewhere. For example, $F_1 = 1 + X_1 X_2$, $F_2 = X_1 X_2$ has a strongly connected graph, but iteration from zero never makes X_2 positive. This degeneracy is excluded. All three conditions are finite structural checks on the monomial list; they hold for (6) and are replayed by the verifier.

6.2 Compactness of bounded multiplicative residuals

Lemma 6.2. *If F is productive, strongly connected, and nonlinear, then, for every $K > 0$, the set*

$$S_K = \{X \in \mathbb{R}_{>0}^d : F(X) \leq KX\}$$

is contained in a compact box $[a, b] \subset \mathbb{R}_{>0}^d$ and is closed in that box. In particular, it is compact, possibly empty.

Proof. Every $X \in S_K$ satisfies $X \geq K^{-1}F(X)$. Iteration from zero of the map $K^{-1}F$ therefore stays below X . Multiplication by a positive scalar does not change which coordinates become positive at any iteration, so productivity supplies a fixed vector $a > 0$ with $X \geq a$ for every $X \in S_K$.

For each edge $i \rightarrow j$, choose a monomial containing X_j . Lower-bound all its other factors, including extra copies of X_j if necessary, by the corresponding coordinates of a . This gives a constant $c_{ij} > 0$ such that

$$F_i(X) \geq c_{ij}X_j \quad (X \geq a).$$

On S_K it follows that $X_i \geq (c_{ij}/K)X_j$. Strong connectivity now gives constants $u_i, v_i > 0$ with

$$u_iX_1 \leq X_i \leq v_iX_1 \quad (X \in S_K).$$

Choose a nonlinear monomial of total degree $q \geq 2$ in row h . The lower comparisons give $F_h(X) \geq cX_1^q$ for some $c > 0$, whereas $F_h(X) \leq KX_h \leq Kv_hX_1$ on S_K . Thus $X_1^{q-1} \leq Kv_h/c$, a uniform upper bound. The other coordinates are then bounded above by their comparisons with X_1 . Finally, the polynomial inequalities are closed within this positive compact box. $\square$

6.3 An elementary convex proof

Put $X_j = e^{y_j}$ and define

$$f_i(y) = \log F_i(e^y) - y_i, \quad h(y) = \max_i f_i(y). \quad (19)$$

Each f_i is a log-sum-exp of affine functions and is convex. For completeness, its Hessian is the covariance matrix of the monomial exponent vectors under their positive softmax probabilities, and is therefore positive semidefinite. Sublevel sets of h correspond to the sets S_K of [lemma 6.2](#). Hence h attains its global minimum.

Theorem 6.3 (Complete rational certificate alternative). *Suppose F has rational coefficients and is productive, strongly connected, and nonlinear. Exactly one of the following holds:*

- (i) *there is $X > 0$ with $F(X) \leq X$;*
- (ii) *there are balanced nonnegative integer monomial weights, not all zero, for which the product (17) is greater than one.*

Here the coefficient $c_{ik}\zeta^{b_{ik}}$ in (17) is simply the corresponding coefficient of F .

Proof. Mutual exclusion is [theorem 5.1](#). Suppose (i) fails. Let y^* minimize h , and put $\delta = h(y^*)$. Since a minimum is attained and no point has $h \leq 0$, we have $\delta > 0$.

Let I be the active rows, those with $f_i(y^*) = \delta$. The origin belongs to the convex hull of their gradients. To see this without invoking an optimization duality theorem, suppose otherwise and let g be the point of that compact convex hull nearest the origin. Then $g \neq 0$ and $g \cdot \nabla f_i(y^*) \geq \|g\|^2$ for every active row. A sufficiently small step in direction $-g$ strictly decreases every active f_i .

The inactive rows remain below the old maximum by continuity and finiteness. This contradicts minimality of h . Consequently there are numbers $\theta_i \geq 0$, supported on I , such that

$$\sum_i \theta_i = 1, \quad \sum_i \theta_i \nabla f_i(y^*) = 0.$$

For each monomial let

$$p_{ik} = \frac{c_{ik} \zeta^{b_{ik}} e^{\alpha_{ik} \cdot y^*}}{F_i(e^{y^*})}, \quad \beta_{ik} = \theta_i p_{ik}.$$

Then $\sum_k \beta_{ik} = \theta_i$, and the gradient identity is exactly

$$\sum_{i,k} \beta_{ik} \alpha_{ik,j} = \theta_j.$$

Thus the real weights β are balanced and have total mass one. Their logarithmic product is

$$\begin{aligned} \mathcal{L}(\beta) &= \sum_{i,k:\beta_{ik}>0} \beta_{ik} \log \left(\frac{c_{ik} \zeta^{b_{ik}} \theta_i}{\beta_{ik}} \right) \\ &= \sum_i \theta_i \log F_i(e^{y^*}) - \sum_j \theta_j y_j^* \\ &= \sum_i \theta_i f_i(y^*) = \delta > 0. \end{aligned}$$

It remains to replace real weights by integers. Restrict to the support of β . On this support, balance and total mass one form a consistent linear system with rational coefficients. Gaussian elimination supplies a rational particular solution and a rational basis for the homogeneous solution space. Approximating the real free parameters by rationals gives balanced rational weights arbitrarily close to β , positive on the same support. The displayed logarithmic expression is continuous there, so a sufficiently close rational approximation still has positive value. Multiplying by a common denominator gives integer weights. Scaling multiplies the logarithmic product by that positive denominator, preserving strict positivity. This proves (ii). $\square$

6.4 What this settles, and what it does not

A positive numerical supersolution bounds the zero iterates and hence all formal partial sums, by the same argument as in [theorem 3.4](#). Thus it forces convergence at the evaluation point. For a proper productive system in the stated class, divergence at a rational point therefore has a finite balanced-monomial obstruction, and a finite obstruction implies the lower-growth bound (18). Thus there is a certificate alternative, not just a test based on watching a floating-point iteration grow. This answers the certificate-existence aspect of Bui's Question 2 for this class, including his seventeen-variable system.

The result does not assert that the original question for every conceivable convolution, maximum, or signed recurrence has been settled. Nor does it give a uniform polynomial bound on the bit length of the weights as a target approaches the exact growth threshold. The explicit certificate below is small and quickly checked; a worst-case certificate-size theory remains a separate problem. The underlying convex mechanism is classical geometric programming, here specialized and proved in an arithmetic certificate form [13, 14, 15].

7 A compact dual certificate for the original recurrence

7.1 The integer weights

For the uncorrected map (6), take

$$\zeta_{\text{dual}} = \frac{100000}{452349}.$$

Use the weights in table 4. There are 53 positive integers, with sum 1,000,000 and largest entry 150,463. Direct integer addition verifies all seventeen balance equations (16).

Row	r_i	Monomial weights, in the order in (6)
c	4,857	3,080, 1,777
d	57,691	29,588, 28,103
e	242,698	93,305, 149,393
f	149,393	88,617, 60,776
g	248,519	150,463, 98,056
h	47,024	27,413, 19,611
p	60,776	39,670, 15,071, 3,712, 1,589, 734
q	113,127	63,384, 36,448, 7,450, 4,066, 1,779
r	6,370	3,666, 2,704
s	27,124	18,469, 6,430, 1,247, 679, 299
t	6,546	3,753, 2,793
u	11,651	7,354, 3,067, 759, 322, 149
v	4,382	2,413, 1,560, 289, 120
w	3,026	2,033, 796, 139, 58
x	8,357	4,786, 2,331, 1,240
y	7,427	4,155, 2,506, 766
z	1,032	702, 256, 74

Table 4: An explicit balanced-monomial certificate. The coefficient of every monomial is one. Within each coordinate row, weights follow the term order in (6).

For example, the incoming C exponent is contributed only by the ξc terms in rows Y and Z . Their weights are 4155 and 702, whose sum is $4857 = r_C$. Other balance equations count repeated variables with multiplicity: the z^2 in row U contributes twice its weight to the incoming Z exponent.

Exact rational interval evaluation proves

$$\frac{1000668023141660979}{10^{18}} \leq \log C(\zeta_{\text{dual}}, m) \leq \frac{1000668023155325499}{10^{18}}. \quad (20)$$

In particular, $1 < \log C < 1001/1000$, so $C > 1$. The slightly generous unit-sized lower margin is merely a convenient verification statement; scaling all weights scales the logarithm.

Theorem 7.1 (Certified limitation of the original system). *The growth μ_{original} of the original equality recurrence satisfies*

$$\frac{452349}{100000} \leq \mu_{\text{original}} \leq \frac{9047}{2000}.$$

Consequently, no supersolution of the original system can establish a growth upper bound smaller than 4.52349.

Proof. Balance and (20) give the lower bound by theorem 5.1 and (18). The repository's rational supersolution at $\zeta = 2000/9047$, replayed by our verifier without a defect correction, gives the upper bound. Finally, a supersolution at a smaller claimed growth value would imply a radius larger than $100000/452349$, contradicting the certified lower bound. $\square$

The last statement can also be read directly: if a supersolution existed at some $\zeta' > \zeta_{\text{dual}}$, monotonicity of Φ in its first argument would make it a supersolution at ζ_{dual} , which the dual certificate forbids. This is an exact obstruction, not a conclusion drawn from a numerical optimizer failing to improve the bound.

7.2 Exact verification without enormous integer products

Computing the rational product (17) literally is possible in principle, but raising many rational factors to large integer weights is unnecessary. Instead, enclose its logarithm using rational arithmetic. For $1 \leq q \leq 2$, let $u = (q - 1)/(q + 1)$, so $0 \leq u \leq 1/3$. The identity

$$\log q = 2 \sum_{k \geq 0} \frac{u^{2k+1}}{2k+1}$$

follows by integrating the geometric series for $2/(1 - u^2)$ from 0 to u . The remainder after M terms satisfies

$$0 \leq \log q - 2 \sum_{k=0}^{M-1} \frac{u^{2k+1}}{2k+1} \leq \frac{2u^{2M+1}}{(2M+1)(1-u^2)}. \quad (21)$$

All endpoints are rational when q is rational.

For a general positive rational q , write $q = 2^e w$ with integer e and $1 \leq w < 2$. Evaluate $\log w$ and $\log 2$ by (21); multiply the interval for $\log 2$ by e , reversing endpoint roles when $e < 0$. This gives a rational enclosure for $\log q$.

The supplied verifier uses $M = 16$. Each individual logarithm interval is rounded outward to denominator 10^{18} before multiplication by its integer weight. This optional outward rounding prevents huge denominators when intervals from many factors are added. It cannot invalidate enclosure. Adding the resulting weighted intervals gives exactly (20). The same cached enclosure of $\log 2$ can be reused; there are at most 54×16 short series terms to evaluate.

The verification cost depends on the bit sizes of the rational monomial factors, weights, and chosen precision, rather than on the number of iterates required to witness numerical divergence. No logarithm supplied by a floating-point library enters the proof. We make no universal claim that certificate discovery or certificate size is polynomial in the reciprocal distance to the growth threshold.

7.3 How the weights were discovered

The following construction is useful when a finite positive critical point has been found numerically. Suppose

$$\tau = \Phi(\rho, \tau), \quad J = \Phi_X(\rho, \tau), \quad \ell^T J = \ell^T, \quad \ell > 0.$$

Normalize by $C = \sum_i \ell_i \tau_i$ and define

$$\beta_{ik} = \frac{\ell_i c_{ik} \rho^{b_{ik}} \tau^{\alpha_{ik}}}{C}.$$

The row sums are $\ell_i \tau_i / C$, and the left-eigenvector identity shows that the weights are balanced. At $\zeta = \rho$, their logarithmic product is zero. At $\zeta > \rho$, it becomes

$$\left(\sum_{i,k} \beta_{ik} b_{ik} \right) \log(\zeta/\rho) > 0$$

provided at least one positively weighted term contains the size variable. Rational approximation inside the balance equations then yields integer weights. This is how the displayed certificate was found.

The approximate critical point and eigenvector are not inputs to the final verifier. Only the resulting integers and rational evaluation point are. The completeness proof in [section 6](#) also does not require that a numerical critical point be correct or even available.

8 The limits of the exact-prefix hierarchy

8.1 A spectral criterion

Fix a positive evaluation point ζ at which the true series $\mathbf{a}$ converges coordinatewise. Put

$$\mathbf{a} = \mathbf{a}(\zeta), \quad p_N = P_N(\zeta), \quad J = \Phi_X(\zeta, \mathbf{a}), \quad J_N = \Phi_X(\zeta, p_N),$$

and define the nonnegative forcing vector

$$b_N = ([\xi^{>N}] \Phi(\xi, P_N(\xi)))(\zeta).$$

At this evaluation point,

$$\Psi_N(\zeta, y) = b_N + \Phi(\zeta, p_N + y) - \Phi(\zeta, p_N). \quad (22)$$

We have $p_N \uparrow \mathbf{a}$ and $J_N \rightarrow J$. Moreover, $b_N \rightarrow 0$: coefficientwise $\Phi(\xi, p_N) \leq \Phi(\xi, \mathbf{a})$, and the latter series converges at ζ . Its high-degree tail bounds b_N .

Theorem 8.1 (Strict stability and obstruction). *Assume the preceding setup.*

- (a) *If $\text{spr}(J) < 1$, then all sufficiently large N admit a real upper certificate at ζ . If the polynomial data and ζ are rational, the certificate can be chosen rational.*
- (b) *For any N , if J_N is irreducible, $\text{spr}(J_N) \geq 1$, and $b_N \neq 0$, then no upper certificate at ζ exists for that N .*
- (c) *If $\text{spr}(J) > 1$, J_N is irreducible for all sufficiently large N , and $b_N \neq 0$ for all sufficiently large N , then no finite-prefix member of the hierarchy admits an upper certificate at ζ .*

Proof. For (a), the nonnegative matrix geometric series gives

$$h = (I - J)^{-1} \mathbf{1} > 0, \quad Jh = h - \mathbf{1}.$$

For sufficiently small $\varepsilon > 0$, polynomial expansion in the nonnegative direction εh gives, uniformly for $0 \leq p_N \leq \mathbf{a}$,

$$\Phi(\zeta, p_N + \varepsilon h) - \Phi(\zeta, p_N) \leq \varepsilon Jh + C\varepsilon^2 \mathbf{1}$$

for some finite $C \geq 0$. Choose ε with $C\varepsilon^2 \leq \varepsilon/4$, and then N sufficiently large that $b_N \leq (\varepsilon/4)\mathbf{1}$. Equation (22) yields

$$\Psi_N(\zeta, \varepsilon h) \leq \varepsilon h - (\varepsilon/2)\mathbf{1}.$$

Thus $p_N + \varepsilon h$ is a certificate with strict slack and strictly positive tail budget. When the data and evaluation point are rational, a sufficiently close rational vector preserves both strict inequalities.

For (b), nonnegative polynomial expansion gives $\Psi_N(\zeta, y) \geq b_N + J_N y$ for $y \geq 0$. Let $\ell > 0$ be a left Perron vector of the irreducible matrix J_N , with eigenvalue $r = \text{spr}(J_N) \geq 1$. A certificate would give $y \geq b_N + J_N y$, and hence

$$(1 - r)\ell^T y \geq \ell^T b_N > 0,$$

which is impossible.

For (c), continuity of spectral radius gives $\text{spr}(J_N) > 1$ for all sufficiently large N . Part (b) rules out certificates at all those levels. By [theorem 3.6](#), an earlier certificate would remain a certificate at all later levels. Therefore no earlier one exists either. $\square$

The equality case $\text{spr}(J) = 1$ is deliberately not classified in general. [Added 29 September 2026, batch 40: Part II classifies it when J is irreducible: then, as for every $\text{spr}(J) \geq 1$, a certificate at ζ exists at level N exactly when the full defect $\Phi(\xi, \mathbf{a}) - \mathbf{a}$ has no coefficients above degree N ([theorem 15.2](#)), so a nonzero full-defect tail excludes every level even when every J_N is subcritical. The reducible critical case remains open ([section 21](#)).] The nonzero forcing and irreducibility conditions in the obstruction are not decorative: without them a critical linear map can have supersolutions. For the Bui map, positive prefix coordinates make the dependency pattern irreducible, and the C row already supplies a nonzero high-degree forcing term whenever the top E coefficient is positive.

8.2 An exact counterexample to asymptotic exactness

Theorem 8.2 (All finite prefixes may retain a growth gap). *There is a scalar proper positive convolution inequality with a true series of growth 1 for which the exact-prefix majorants have growth tending to 3, not 1.*

Proof. Take

$$A(\xi) = \frac{\xi}{1-\xi} = \sum_{n \geq 1} \xi^n, \quad \Phi(\xi, x) = \xi + x^2.$$

The coefficient inequality $A \leq \xi + A^2$ holds: the coefficient of ξ^n in A^2 is $n-1$ for $n \geq 2$. The exact-prefix defect is

$$D_N(\xi) = \sum_{n=3}^N (n-2)\xi^n,$$

with an empty sum for $N < 3$. The corrected equality equation is

$$B_N = \xi + B_N^2 - D_N(\xi). \tag{23}$$

Let $c_N(t) = t - D_N(t)$ for real t . Since

$$D_\infty(t) = \frac{t^3}{(1-t)^2},$$

we have $D_\infty(1/3) = 1/12$ and $D'_\infty(t) \leq 1$ on $0 \leq t \leq 1/3$, with equality only at the right endpoint. A finite truncation has strictly smaller derivative there. Thus c_N is strictly increasing on this interval, $c_N(1/4) \leq 1/4$, and $c_N(1/3) > 1/4$. There is a unique

$$\rho_N \in [1/4, 1/3) \quad \text{with} \quad c_N(\rho_N) = 1/4.$$

At $t = \rho_N$, the value $v = 1/2$ solves (23), and

$$P_N(\rho_N) < A(\rho_N) < 1/2.$$

It is therefore a valid upper certificate. For any $t \in (\rho_N, 1/3]$, the discriminant of $x^2 - x + c_N(t) = 0$ is negative, so there is no finite real fixed point. Hence the radius of B_N is exactly ρ_N .

The truncated polynomials D_N converge uniformly to D_∞ on $[0, 1/3]$. Every limit point ρ of the sequence ρ_N therefore satisfies

$$\rho - \frac{\rho^3}{(1-\rho)^2} = \frac{1}{4}, \quad \text{equivalently} \quad (3\rho - 1)^2 = 0.$$

Thus $\rho_N \rightarrow 1/3$, and the corrected growth tends to 3. The true series has radius 1 and growth 1. $\square$

The same barrier is visible directly in [theorem 8.1](#):

$$\Phi_x(t, A(t)) = \frac{2t}{1-t}$$

crosses one at $t = 1/3$, well before the true singularity at $t = 1$. Every finite coefficient of B_N eventually agrees with A , but radii of convergence do not pass continuously through this coefficientwise limit. [Added 29 September 2026, batch 40: Part II proves that this is the general mechanism ([theorem 16.3](#)) and gives the rate for this example, $\frac{1}{3} - \rho_N \sim \frac{2}{9} \sqrt{2N-1} 3^{-N/2}$ ([section 20](#)).]

This counterexample is not a disproof of Bui's separate Question 1 about increasing neighborhood size with improved geometric recurrences. Here the local polynomial map is held fixed and only a finite exact prefix is substituted. Those are different hierarchies. It does show that an eventual convergence claim for the present prefix method requires a stability argument or additional geometric refinement.

8.3 A conditional sensitivity formula for choosing improvements

A useful guide for future searches is available when a smooth critical branch has already been justified. Suppose an admissible defect improvement $D + \varepsilon E$ has critical data $(\rho(\varepsilon), \tau(\varepsilon))$ satisfying

$$\Phi(\rho(\varepsilon), \tau(\varepsilon)) - D(\rho(\varepsilon)) - \varepsilon E(\rho(\varepsilon)) = \tau(\varepsilon).$$

At $\varepsilon = 0$, let $\ell^T J = \ell^T$ for $J = \Phi_X(\rho, \tau)$. Differentiating and multiplying by ℓ^T cancels the unknown derivative of τ , yielding

$$\rho'(0) = \frac{\ell^T E(\rho)}{\ell^T (\Phi_\xi(\rho, \tau) - D'(\rho))}, \quad (24)$$

provided the denominator is nonzero. For a positive tail formulation with $\tau = P(\rho) + y$, the denominator also equals $\ell^T \Psi_\xi(\rho, y)$, because the difference is $\ell^T (J - I)P'(\rho) = 0$. It is nonnegative, and is positive when some size-dependent monomial contributes positively.

Formula (24) suggests prioritizing verified defect improvements by their Perron-weighted value. It is conditional on the stated smooth critical family; it is not used in [theorem 4.1](#). In particular, it never authorizes subtracting an unproved or merely guessed defect.

9 Verification, formalization, and reproducibility

9.1 The proof dependency chain

The mathematical chain behind the new upper bound is

$$\begin{aligned} & \text{geometric recurrence lemma} + \text{exact marked prefix} \\ & \implies \text{nonnegative corrected tail system} \\ & \implies \text{rational supersolution at } 500/2249 \\ & \implies \text{coefficient majorant} \implies \lambda \leq 2249/500. \end{aligned}$$

The new lower certificate for the old equality system follows a different chain:

$$\begin{aligned} & \text{integer balance} + \text{rational log enclosure} \\ & \implies \text{no positive supersolution} \\ & \implies \mu_{\text{original}} \geq 452349/100000. \end{aligned}$$

These chains share the monomial map but not the logical direction of the majorant comparison.

9.2 Archive contents and commands

The archive includes the article source and PDF, the exact marked profile, twelve upper certificates, the dual certificate, the verification transcript, and source code. The principal files are:

File	Purpose
code/model.py	Sparse map, patterns, exact polynomial operations
code/verify.py	Standard-library rational certificate replay
code/enumerate.cpp	Incremental marked enumeration through size 18
code/enumerate_direct.cpp	Direct-scanning comparison implementation
code/research.py	Independent Python enumeration and numerical exploration
code/discover_certificates.py	Numerical search and rationalization
data/profiles.json	Exact marked and unmarked counts
data/upper_N18.json	Main rational upper certificate
data/dual_original.json	Balanced lower certificate for original model
data/verification.json	Exact arithmetic residuals and log interval

From the archive root, the fast arithmetic audit is

```
python3 code/verify.py --report data/verification.json
```

It uses only Python’s standard library, arbitrary-precision integers, and `fractions.Fraction`. It checks the monomial graph hypotheses, nonnegative prefix defects, five occupancy partitions, all upper-certificate residuals and tail budgets, all dual balance equations, the rational log interval, and the repository’s previous 4.5235 supersolution.

Full regeneration of the main finite table is separate:

```
c++ -O3 -std=c++17 code/enumerate.cpp -o enumerate
./enumerate 18 > regenerated.csv
python3 code/check_profile.py regenerated.csv
```

This visits more than two billion partial objects in total and is much slower than certificate replay. The comparison tool checks all marked and unmarked entries against the archived table. No external sequence database is needed for either enumeration or certificate checking.

The numerical discovery scripts require NumPy, SciPy, and SymPy. They are provided for further experimentation, but are outside the proof boundary. The figures in the numerical-estimate column of [table 3](#) should not be promoted to exact assertions merely because those scripts print a small residual or a successful solver status.

[write] *Layout in ProveIt.* The report directory keeps the delivered code/, data/ and tex/ layout, so the commands above run from the report root under the shipped names. Two files moved: the delivered Makefile is code/Makefile (its recipes still use root-relative paths, so run it as `make -f code/Makefile verify` from the report root), and `requirements-discovery.txt` is data/requirements-discovery.txt. The delivered PDF was replaced by a build of this text, and the delivered checksum manifest is not shipped. On the intake machine python3 is spelled py. Because `verify.py -report` rewrites data/verification.json, `make_tables.py` rewrites tex/*.tex, and `discover_certificates.py` overwrites data/upper_N*.json and data/dual_original.json, run them on a copy of the directory. At intake, `verify.py` passed on such a copy and its output equalled data/verification.txt.

9.3 A focused Lean extension plan

The most direct formal extension reuses `geometricPublishedBuiRecurrences` and the existing domination of the fixed-polyomino count by the G coordinate. The genuinely new interfaces are

an exact finite prefix for the marked profile, the defect-corrected weighted-prefix inequality, and the new rational vector.

A practical route can avoid formalizing complex singularity theory entirely. First prove the finite convolution identity underlying (9). Next prove a weighted-tail version of the repository’s monotone-prefix certificate theorem. Instantiate it with the new vector, using ordinary exact rational reduction for every inequality. The existing growth endpoint can then carry the pointwise exponential bound to λ . The expensive part is a formally verified enumeration or a small proof object certifying the marked prefix, not the seventeen final inequalities. [write] In the Lean files at the pin, the “monotone-prefix certificate theorem” exists in ζ -generic form only as `buiIterate_le_supersolution` and `PrefixRecurrence.le_supersolution`; the endpoint theorems that carry a supersolution to a pointwise bound are specialized to $\zeta = 2000/9047$ (section 1.5). The final passage to λ can use the generic `growthSup_le_of_le_pow` (`Growth.lean:38`), since $v_G < 1$ gives $A_n \leq (2249/500)^n$.

For dual certificates, the balance identity is finite integer arithmetic. One can either formalize weighted arithmetic–geometric mean directly or expand rational weights into an unweighted finite product argument. A logarithmic checker with the remainder bound (21) avoids large powers and is a natural reusable component. The abstract completeness theorem need not be formalized before the particular 53-weight certificate can be checked.

No Lean source claiming these new results is included. In particular, the presence of a Lean proof of the old bound in the repository must not be confused with a Lean proof of the new finite-prefix theorem or of the size-18 data.

10 Proposed research questions and next projects

The following projects are intended as concrete continuations, not claims already established in this report.

1. **Kernel-check the 4.498 bound.** Formalize the finite-prefix certificate lemma and a verified marked polyomino enumerator, or design a compact reflective certificate for its output. The target should be an unconditional endpoint with precisely the same counting conventions as `GeometricComplete.lean`. This is the highest-priority validation project. [write] Section 1.5 lists the Lean pieces that exist, the ζ -specialization that must be generalized, and a smaller first target (the $N = 7$ certificate, $\lambda \leq 4.5233$, needs counts only through size 7).
2. **Generate longer marked prefixes without listing every object.** Adapt a transfer-matrix or connectivity-state enumeration to accumulate all seventeen occurrence counts, not merely A_n . The relevant challenge is preserving marked local information while merging equivalent boundary states. Exact counts of larger objects could strengthen the present bound without changing the imported geometric proof.
3. **Certify lower bounds for the corrected envelopes.** Expand or represent the positive maps Ψ_N and construct balanced-monomial certificates for them. This would replace the middle column of table 3 by narrow rigorous two-sided brackets. Sparsity or an arithmetic-circuit representation may avoid a large explicit expansion.
4. **Locate the actual stability barrier of the fixed Bui map.** Bound $\text{spr}(\Phi_X(\zeta, a(\zeta)))$ using certified lower and upper bounds on the true marked generating functions. Determine whether the prefix hierarchy’s limiting obstruction lies substantially below the true polyomino radius. The strict criteria in theorem 8.1 give concrete sufficient tests, while the equality case remains open. [Added 29 September 2026, batch 40: Part II closes the irreducible equality case (theorem 15.2) and proves that the limiting majorant growth of the fixed map is at least 4.3149 (theorem 22.2); where the barrier actually lies, and whether it is below the true radius,

remain open (section 24).]

5. **Optimize which geometric deficiencies to repair.** Use Perron-weighted sensitivities as a search heuristic, but accept a repair only after proving the corresponding injection or coefficient improvement. A small new neighborhood resolving a heavily weighted overcount may be more valuable than many additional unweighted states.
6. **Study Bui’s neighborhood-refinement convergence question.** Develop a hierarchy in which neighborhood size and geometric recurrences are refined together. Prove convergence to the true growth constant, or identify a rigorous obstruction. The scalar counterexample here rules out one naive prefix-only argument; it does not settle the neighborhood question.
7. **Quantify certificate bit complexity.** Given a rational target separated from the critical value by a known amount, bound the sizes of balanced integer weights and the number of logarithm terms required. Separate complexity of discovery from complexity of replay. The present million-unit total weight is an example, not a worst-case bound.
8. **Extend the alternative beyond the irreducible nonlinear case.** Decompose reducible systems into strongly connected blocks and analyze linear, unproductive, and boundary blocks separately. A comprehensive theorem should explain when weak infeasibility can occur and when a strict rational obstruction still exists. Maximum operators require a different or enriched certificate language.
9. **Obtain certified critical asymptotics.** For selected corrected systems, prove uniqueness of the relevant positive critical point, nondegeneracy, and the resulting coefficient asymptotics. Such a result would justify numerical critical continuation as mathematics, not merely as a source of candidate certificates. [Added 29 September 2026, batch 40: Part II proves general radius, height and local-amplitude asymptotics as $N \rightarrow \infty$ at an interior nonlinear crossing (theorems 18.2 and 19.1). It proves no coefficient asymptotics and does not verify its hypotheses for the Bui system, so this project stays open.]
10. **Transfer the method to other lattice animals and local grammars.** Identify an existing positive recurrence system on a different lattice, compute its first exact defects, and test whether a modest prefix correction produces a useful improvement. The general theorems apply without change once properness and the underlying combinatorial domination are proved.

11 Conclusion

The repository’s polyomino project supports more than a small numerical retuning. Incorporating exact marked prefixes changes the majorant itself, and the resulting rational certificate proves $\lambda \leq 4.498$. Balanced monomial weights provide the missing lower-certificate mechanism for a natural class of convolution systems; their explicit use also proves that the original recurrence cannot support the new bound without a structural correction. Finally, the spectral analysis identifies why this successful improvement must not be extrapolated automatically to asymptotic exactness. The central next step is independent verification, especially formal certification of the finite marked enumeration. [Added 29 September 2026, batch 40: Part II, which follows the appendices of Part I, makes that limitation exact.]

A Complete marked profiles through size eighteen

The tables give the exact occurrence counts for the patterns in [table 1](#). All omitted size-zero entries are zero. The row A_n counts translation classes; every other row counts marked occurrences. These data, together with (6), determine every coefficient of D_{18} by finite integer convolution. The JSON and CSV forms contain the same integers without typographical formatting.

n	1	2	3	4	5	6	7	8	9
A_n	1	2	6	19	63	216	760	2725	9910
C	1	1	1	3	10	35	128	479	1819
D	1	1	2	6	20	71	260	973	3696
E	1	1	3	10	35	128	479	1821	7001
F	1	3	10	35	128	481	1841	7131	27856
G	1	2	6	20	71	261	982	3751	14481
H	1	2	5	15	51	182	670	2517	9592
P	0	1	4	15	57	220	859	3380	13375
Q	0	1	3	10	36	133	503	1930	7480
R	0	1	3	8	25	85	303	1110	4150
S	0	1	3	9	31	111	410	1544	5896
T	0	1	3	9	30	106	388	1451	5510
U	0	1	4	14	50	185	700	2686	10412
V	0	0	1	4	14	52	196	748	2884
W	0	0	1	4	13	46	168	625	2362
X	0	1	2	5	16	54	192	703	2626
Y	0	1	2	4	12	39	135	485	1788
Z	0	1	2	3	8	25	84	296	1080

Table 5: Exact profiles for sizes 1 through 9.

n	10	11	12	13	14	15	16	17	18
A_n	36446	135268	505861	1903890	7204874	27394666	104592937	400795844	1540820542
C	6979	26988	105011	410629	1612207	6351215	25091789	99372469	394381157
D	14190	54917	213865	837024	3289287	12969734	51285522	203287656	807484165
E	27144	105927	415489	1636445	6467194	25630675	101822518	405337049	1616425239
F	109503	432550	1715129	6821554	27199077	108673645	434959946	1743471996	6997283445
G	56348	220593	867716	3426361	13572506	53905206	214569185	855714279	3418228578
H	36929	143269	559155	2192740	8632427	34094245	135023622	535974728	2131792166
P	53155	211957	847413	3395193	13626571	54768439	220390761	887757717	3579054867
Q	29204	114666	452227	1789916	7105312	28274531	112746667	450377230	1801803339
R	15744	60366	233323	907529	3547802	13926810	54857224	216706457	858185665
S	22739	88352	345290	1355716	5343140	21124511	83738100	332687072	1324308001
T	21162	81966	319543	1252166	4927255	19455297	77039780	305805462	1216400095
U	40678	159874	631249	2501630	9943629	39622046	158205688	632793092	2534808294
V	11212	43866	172467	680772	2695907	10704713	42601646	169868638	678451341
W	9035	34875	135550	529733	2079362	8191718	32369106	128231694	509107371
X	9950	38100	147076	571394	2231348	8750584	34438134	135936824	537948754
Y	6709	25491	97773	377796	1468440	5735092	22488118	88474763	349078294
Z	4025	15214	58108	223734	866978	3377002	13210216	51861498	204221945

Table 6: Exact profiles for sizes 10 through 18.

B A minimal exact certificate checker

The following core is sufficient to check an upper candidate once the map and true prefix are supplied. The full verifier also checks the input conventions, all intermediate certificates, the dual weights, and its rational logarithm enclosure.

```

from fractions import Fraction as Q

# P[i] and D[i] are integer coefficient lists, low degree first.
def evaluate(coefficients, zeta):
    value = Q(0)
    for coefficient in reversed(coefficients):
        value = value * zeta + coefficient
    return value

def check_upper(zeta, values, P, D, polynomial_map):
    prefix = [evaluate(p, zeta) for p in P]
    defect = [evaluate(d, zeta) for d in D]
    image = polynomial_map(zeta, values)
    tail = [v - p for v, p in zip(values, prefix)]
    residual = [v - f + d
                 for v, f, d in zip(values, image, defect)]
    if any(t < 0 for t in tail):
        raise ValueError("Candidate lies below the known prefix")
    if any(r < 0 for r in residual):
        raise ValueError("A supersolution inequality failed")
    return tail, residual

```

The defect is computed, not guessed. If $P_i(\xi) = \sum_{n=1}^N a_{i,n} \xi^n$, then for each row i one expands the finitely many monomials in $\Phi_i(\xi, P)$, discards degrees above N , and subtracts P_i . Because all numbers involved are integers before evaluation, this step is particularly suitable for independent implementations or proof-assistant reflection.

For the dual checker, (16) is a sum of integer vectors. The j th entry of a monomial exponent vector is counted with its full multiplicity. Omitting the factor two for a square, for example, would invalidate the certificate. The supplied verifier iterates over repeated variable indices, so this multiplicity is explicit in its sparse format.

C Source audit and claim classification

Claim or ingredient	Status and evidence
Geometric inequalities	Imported from Bui v2, Lemma 4 and Appendix B; corresponding repository endpoint source inspected at the pinned commit. [write] In ProveIt a Lean theorem for the actual counts, <code>geometricPublishedBuiRecurrences</code> (section 1.5).
Old 4.5235 upper bound	Existing repository result; rational vector independently replayed in the new verifier. No new Lean build was performed. [write] Kernel-checked in ProveIt; the project's Lean-verified bound.
Prefix-corrected majorant	Proved in section 3, including the nonnegative tail map and the known-prefix lower-budget condition.
New 4.498 bound	Computer-assisted theorem: exact marked enumeration plus a rational certificate, with the detailed dependency chain in section 9.

Claim or ingredient	Status and evidence
Marked table through 18	Generated by exhaustive C++ enumeration. Python agrees through 11; direct and incremental C++ scans agree through 16; standard unmarked totals agree through 18. Not kernel-checked. [write] Regenerated through 18 at intake from the final source, and through 10 by an independent enumerator (section D).
[write] New 4.498 bound in Lean	Not formalized; the Lean endpoints are specialized to $\zeta = 2000/9047$ (section 1.5).
Dual obstruction	Elementary weighted AM–GM proof; particular integer balance and strict log inequality replayed exactly.
Certificate completeness	Proved for productive, strongly connected, nonlinear positive polynomial systems with rational data. Not claimed for all maximum or signed recurrences. Based on classical geometric-programming ideas.
Numerical critical values	Exploratory estimates only; not certified optimal values and not used to establish the accepted inequalities.
Stability theorem and gap	Written proofs in section 8 . The case of limiting spectral radius exactly one is not classified generally. [Added 29 September 2026, batch 40: classified for irreducible J in Part II, theorem 15.2 ; written proof, not formalized.]
Sensitivity formula	Conditional on an already justified differentiable critical family. Not used for the headline bound.
Novelty	The explicit bound and application were not found in the inspected sources. No claim of an exhaustive priority search or independent refereeing.

D [write] Provenance, intake verification and non-claims

D.1 Provenance

This report was built from one manuscript: manuscript 06 of ProveIt’s incoming-reports batch 36, the archive ProveIt_Polyomino_Research (delivered in commit e13affd32, placed in commit 1a1396d4d). It was written against the pin e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9; the project it continues is unchanged between that pin and the placement commit, so its statements about the project are current except where the delivered text undersells the formal status of the geometric lemma ([section 1.5](#)). As a single source it required no merge choices. Its whole text is printed; passages marked [write] were added when it was written into ProveIt: [sections 1.4](#) and [1.5](#), this appendix, the paragraph on earlier upper bounds in [section 1](#), notes in the title-page box and the “three levels” box, after (8) and the geometric-lemma paragraph of [section 2](#), on intake regeneration in [section 4](#), on layout and on the Lean plan in [section 9](#), after research question 1, three rows of the audit table in [section C](#), and four bibliography entries ([7, 8, 9, 10]). All labels carry the prefix kfp: except eq:map, which the delivered, generated tex/dual_table.tex cites by that name.

[Added 29 September 2026, batch 40:] The report now has a second source. Batch-40 manuscript 05 (archive ProveIt_Spectral_Saturation, arrived in 017e6c0d8, placed in afb2d1227, pinned to 74f7f5bdb) is printed as Part II, with its own provenance in [section 12](#); its labels carry the prefix kfp:sat:. The Part I text above changed only by dated “Added 29 September 2026” pointers, the entries for Part I and its appendices in the contents, the label kfp:sec:conclusion, four macros for Part II in the preamble, and the PDF subject line.

D.2 Intake verification

- The arithmetic verifier code/verify.py passed on a copy of the package (4.3 s) and printed exactly data/verification.txt; the profile checker accepted data/profiles_18.csv.
- The size-18 table was regenerated from scratch by a fresh build of code/enumerate.cpp and matched in every entry (section 4); fresh runs through sizes 11–14, 16 and 17 and a direct-scan run through 12 matched too.
- An independent set-based Python enumerator, with coordinates from the project's Patterns.lean, matched all entries through size 10. An independent transcription of (6) from the project's research note [7] reproduced (13), the size-18 margins and v_G , and confirmed the old 4.5235 vector as a supersolution.
- The seventeen offset sets of table 1 were compared entry by entry with Patterns.lean, and the unmarked column with OEIS A001168 through 18.
- A 50-digit evaluation of $\log C(\zeta_{\text{dual}}, m)$ gave 1.00066802314520877... , inside (20).

These checks do not verify the written proofs, and no independent proof review was performed. At intake the proofs of lemma 3.2, theorems 3.3, 3.4, 3.6, 5.1, 6.3 and 8.2 were checked and that of theorem 8.1 was read; no gap was found.

D.3 What this report does not claim

Every limitation stated by the delivered text is kept; collected:

- N1. The lower bound $\mu_{\text{original}} \geq 452349/100000$ concerns the uncorrected equality recurrence and is *not* a lower bound on λ (section 5).
- N2. No new Lean formalization. The Lean proof of 4.5235 in ProveIt is not a proof of 4.498, of the finite-prefix theorem, or of the size-18 data.
- N3. The arithmetic checker does not prove that the table enumerates the stated objects; its partition, defect and G -domination checks are necessary conditions only.
- N4. The enumeration cross-checks are not a formal certificate that every size-18 object was generated exactly once. The delivered record made no fresh size-18 run after the final command-line changes; the intake regeneration closes that gap only with the same program, and the two C++ implementations share one generation strategy.
- N5. The report is an AI-assisted draft and has not been independently refereed.
- N6. Novelty is assessed only against the inspected repository snapshot and sources; there was no exhaustive priority search.
- N7. Geometric programming, weighted AM–GM and Redelmeier-style enumeration are prior tools, not claimed inventions.
- N8. The certificate alternative is proved only for productive, strongly connected, nonlinear systems with rational data; it gives no uniform bound on certificate bit length and does not cover maximum or signed recurrences.
- N9. The case $\text{spr}(J) = 1$ of theorem 8.1 is not classified. [Added 29 September 2026, batch 40: re-scoped. Part II classifies it for irreducible J (theorem 15.2); the reducible case is still not classified.]
- N10. The counterexample of theorem 8.2 does not disprove Bui's Question 1.

N11. The sensitivity formula (24) is conditional and not used in [theorem 4.1](#).

N12. The numerical critical estimates in [table 3](#) are not certified values or bounds.

N13. The OEIS comparison is a consistency check, not an input to the proof or the checker.

Added in the write: (W1) the geometric lemma is kernel-checked in `ProveIt`, but only relative to the transcription of Bui’s diagrams into `Patterns.lean`; (W2) the bound 4.498 remains unverified by `Lean`, and the project’s `Lean`-verified bound remains 9047/2000; (W3) the earlier bounds of [section 1](#) were not re-examined, and their sources were not read beyond their bibliographic data.

Part II. Spectral saturation of the exact-prefix hierarchy

Sections 1–11 and Appendices A–D form the original report (Part I), written into ProveIt in batch 36, unchanged except for dated pointers to this addition. Part II was added on 29 September 2026 in batch 40 of ProveIt’s incoming-report intake, from one manuscript written against the state of Part I printed here. It classifies the equality case $\text{spr}(J) = 1$ that Part I left unclassified after [theorem 8.1](#) (limitation N9 of [section D](#)), for an irreducible true-profile Jacobian only, determines the limiting radius of the whole prefix hierarchy, and proves that no exact prefix can take the fixed Bui map’s majorant growth below 4.3149. [Section 12](#) records the provenance and fixes the notation of the addition.

12 Provenance, conventions and claim boundary of Part II

12.1 One manuscript, one addition

Part II is batch-40 manuscript 05, *When Exact Prefixes Stop Improving: Spectral saturation, sharp square-root laws, and a certified barrier for convolution majorants*, an AI-assisted mathematical draft prepared for Vladimir Reshetnikov and dated 28 September 2026 (27 pages as delivered). It arrived as the archive `ProveIt_Spectral_Saturation.zip` (inner directory `ProveIt_Spectral_Saturation/`) in ProveIt commit `017e6c0d8` and was placed in `afb2d1227`. It was written against ProveIt commit `74f7f5bdb`, at which Part I’s `article.tex` had the Git blob `4a021fb3`. That blob is unchanged at the placement commit, and so are the project `Combinatorics/Polyominoes/KlarnnerConstant` and Part I’s `code/model.py` and `data/profiles_18.csv`; every statement the manuscript makes about Part I therefore refers to the text printed above.

Its code, data and audit files are shipped beside Part I’s with the prefix `02-spectral-saturation-` (Part I’s unprefixed files count as 01): three Python scripts and a build script in `code/`, five data files in `data/`, and `02-spectral-saturation-PROOF_STATUS.md` and `02-spectral-saturation-SOURCES.md` at the root, all byte-identical to the delivery. Its manuscript, README and PDF are not shipped and survive in the arrival commit. Its `data/profiles_18.csv` is not shipped either: the manuscript copied it from Part I, and it is byte-identical to Part I’s `data/profiles_18.csv` (blob `82648328`), which is the file every mention of it in Part II now means.

The manuscript contributes, in the order printed here: its own introduction and claim boundary ([section 13](#)); a restatement of the prefix construction ([section 14](#)); the critical alternative, which classifies the irreducible case $\text{spr}(J) \geq 1$ ([section 15](#)); the exact limiting radius of the hierarchy ([section 16](#)); a tail estimate without coefficient regularity ([section 17](#)); the square-root law at an interior nonlinear crossing ([section 18](#)); the scaled branch and the quarter-power amplitude ([section 19](#)); sharp equivalents for Part I’s scalar example ([section 20](#)); four examples that delimit the hypotheses ([section 21](#)); the all-prefix 4.3149 barrier for the fixed Bui map ([section 22](#)); its reproducibility and proof audit ([section 23](#)); ten research projects ([section 24](#)); its conclusion; and, as the last three sections of Part II, its three appendices (the map, a replay algorithm, and the source ledger). Every result, proof, example, remark, limitation and question of the manuscript is printed. Its bibliography is merged into the report’s; the entries it shares with Part I (Bui’s paper) are printed once.

Material already in Part I. The manuscript restates, to be self-contained, Part I’s formal comparison lemma, prefix-corrected majorant, certificate theorem, monotonicity, strict subcritical criterion and scalar limit 3, and says so itself. These restatements are kept where Part II’s proofs cite them in its own formulation, each followed by a [write] note naming the Part I result; where

the manuscript's proof differs from Part I's it is kept as a second proof. The manuscript's display of the 53-monomial map duplicates (6) and is printed once, there (section 26).

The manuscript's abstract.

Exact initial coefficients can strengthen a positive convolution recurrence without changing its combinatorial proof. They need not, however, make its majorant asymptotically exact. We study this limitation for a proper nonnegative polynomial system $A \leq \Phi(\xi, A)$, its defect $D = \Phi(\xi, A) - A$, and the corrected equations $B_N = \Phi(\xi, B_N) - [\xi^{\leq N}]D$.

First, at any convergent evaluation where the true-profile Jacobian is irreducible and has spectral radius at least one, a level- N upper certificate exists *if and only if* the defect tail vanishes. This settles the irreducible critical case left open in the repository's finite-prefix report. It also identifies the exact limiting radius of the hierarchy: the first Perron crossing, or the true radius if no earlier crossing occurs.

Second, at an interior nonlinear crossing r_* , put $\eta_N = \ell^T(D - [\xi^{\leq N}]D)(r_*)$. With explicitly defined positive constants κ and β , we prove

$$r_* - \rho_N \sim \frac{2\sqrt{\beta}}{\kappa} \sqrt{\eta_N}, \quad B_N(\rho_N) - A(\rho_N) \sim u \sqrt{\eta_N/\beta}.$$

A universal scaled branch and a quarter-power law for its local singular amplitude follow. No regular variation or nonlacunarity assumption on the defect coefficients is needed.

Finally, using the repository's size-18 marked polyomino profile, an exact rational vector proves that every member of its fixed-map prefix hierarchy has growth at least 4.3149. This is a lower bound on the *majorants*, not on polyomino growth. The package includes a standard-library arithmetic checker, exact scalar root brackets, source provenance, and a research agenda. The proofs are unrefereed and not Lean-verified; the size-18 enumeration is inherited rather than rerun.

("The repository's finite-prefix report" is Part I. The size-18 table was not rerun for the manuscript, but Part I's intake regenerated it from the final enumerator source; see section D.)

Status of Part II. Its analytic theorems have written proofs; it is an AI-assisted, unrefereed draft, and independent review of the uniform fold argument (lemma 18.3 and section 18) is recommended before correctness or novelty is treated as settled. Novelty is a bounded assessment against the sources inspected, not a priority claim. **Nothing in Part II is formalized** in any proof assistant; the project's Lean-verified bound remains $\lambda \leq 9047/2000 = 4.5235$ (section 1.5). Its arithmetic checker proves finitely many rational inequalities, not the analytic theorems; its size-18 input is Part I's enumeration, which is not kernel-checked.

What 4.3149 is. Theorem 22.2 proves that every exact-prefix majorant of the fixed seventeen-variable Bui map has growth $\mu_N \geq 4.3149$: a limit of this *method*. *Tempting false reading:* " $\lambda \geq 4.3149$ ", a lower bound on Klarner's constant. It is not; the true counts lie *below* the majorants. *Tempting false reading:* "4.3149 improves on 4.498". It does not; it is not an upper bound at all, but a floor under every upper bound this method can certify. Part I's 4.498 (computer-assisted) and the Lean-verified 4.5235 stand unchanged above it.

What N9 now is. The equality case $\text{spr}(J) = 1$ is classified for *irreducible* J (theorem 15.2); the reducible critical case, the crossing at the true singularity, and the exact fixed-map saturation constant remain open (sections 21 and 24).

12.2 Notation of Part II

Part II keeps the manuscript's notation with two changes, made so that it reads consistently with Part I:

- the manuscript's formal size variable z (its Sections 1–8) is written ξ , as in Part I and as the manuscript itself does from its Section 9 on, where z is a coordinate of (6);
- the manuscript's Perron eigenvalue λ , $\lambda(t) = \text{spr}(J(t))$ (proof of [theorem 15.2](#), [lemma 18.1](#) with (42), [lemma 18.3](#)), is written Λ , because λ is Klarner's constant in Part I.

No normalization changed. Macro-level spellings (the manuscript's $\mathcal{R}_N$, $[z^{\leq N}]$, $[z^{>N}]$, $\mathbf{1}$) are typeset as $\mathcal{R}_N$, $[\xi^{\leq N}]$, $[\xi^{>N}]$, $\mathbf{1}$. The delivered, generated table data/02-spectral-saturation-scalar_table.tex ([table 9](#)) heads its gap column $\rho_* - \rho_N$; that ρ_* is the crossing r_* of the text, here 1/3.

Table 8: Symbols of Part II against Part I.

Symbol	Meaning in Part II	In Part I
ξ	formal size variable (the manuscript's z in its Sections 1–8)	the same
A, A_i	the true series (a general subsolution $A \leq \Phi(\xi, A)$)	a ; A_n is the polyomino count
t, t_0	a positive evaluation point; $t_0 = 10000/43149$	ζ ; $\zeta = 500/2249$ for 4.498, ζ_{dual}
a	$A(t)$; also the constants of lemma 17.1 ($0 < a < r < b$) and of the fold argument, and a_N the singular amplitude	$a = a(\zeta)$ in section 8 ; $a_{i,n}$ counts
D, D_N	full defect $\Phi(\xi, A) - A$; $D_N = [\xi^{\leq N}]D$	D_N the same; D also the defect polynomial of (24); D a coordinate
$\mathcal{R}_N$	the defect tail $D - D_N$	$[\xi^{>N}] \cdot$ is the operator $[\xi^{>N}]$
$b_N(t)$	$([\xi^{>N}]\Phi(\xi, P_N))(t)$, finite forcing	the same b_N (section 8); not the project's b_N
P_N, Y_N, B_N, Ψ_N	prefix, tail, majorant, tail map	the same
$J, J(t), J_*$	Φ_X at the true profile $A(t)$; J_* at r_*	J the same; J_N at the prefix
$\Lambda, \Lambda(t)$	Perron value of $J, J(t)$ (the manuscript's λ)	λ is Klarner's constant
R	system radius of A	required offset set; radii R_i
ρ_N, μ_N	system radius of B_N , $\mu_N = 1/\rho_N$	ρ_N as in theorem 8.2 ; μ_{original} is the growth of the uncorrected recurrence
r_*	first Perron crossing (36)	—
ℓ, u	left and right Perron vectors, $\ell^T u = 1$	ℓ as in (24); u a coordinate
$\kappa, \beta, \eta_N, \varepsilon_N$	invariants (40); $\varepsilon_N = \sqrt{\eta_N}$ (also $\sqrt{H_N(r)}$ in lemma 17.1)	ε a small parameter
h	$v - a$ or $B_N(t) - A(t)$	$h = (I - J)^{-1}\mathbf{1}$
$Q_t(h)$	nonlinear remainder (32)	Q a coordinate
$G(t, h, r)$	the map (49)	G the coordinate whose counts dominate A_n
F_N, F	scaled reduced function (52); F also a generic vector series	$F(X) = \Phi(\zeta, X)$; F a coordinate
$K, K(t)$	scaled distance δ/ε_N and a bound in lemma 17.1 ; $K(t) = \sum_{k < d} J(t)^k$	—
S, S_c, S_-	a subsolution (lemma 14.1); scaled height	sets S, S_K ; S a coordinate

Symbol	Meaning in Part II	In Part I
H, H_N	a scalar series and its tail (lemma 17.1); $H_N(t)$ also the scalar sign oracle (section 27)	H the nonlinear part of Φ
$T(\xi, Y)$ w	a generic proper map witness vector (table 10); direction in proposition 16.1 ; complement component $h - su$	$T_N = [\xi^{\leq N}] \Phi(\xi, P_N)$ coordinate w ; $w = v - P_N(\zeta)$
r, r_N	a point, a vector parameter in (49), an index; $r_N = \ \mathcal{R}_N(t)\ _\infty$	r a coordinate; row sums r_i
γ	1000001/1000000	—
e_M	the e -coefficient of size M	$a_{E,M}$

13 The question, the answer, and the claim boundary

13.1 Why a better prefix may still be the wrong limit

Suppose a combinatorial counting problem gives a vector of generating functions $A(\xi)$ and a componentwise upper recurrence

$$A(\xi) \leq \Phi(\xi, A(\xi)),$$

where Φ is a polynomial with nonnegative coefficients. The equality solution is an upper bound, but its recurrence can overcount. A natural repair is to compute exact coefficients through size N and subtract the corresponding overcounts. The resulting majorant agrees with the true series through size N and improves as N increases.

There are two very different possible conclusions. One is that every fixed coefficient eventually becomes correct. The other is that the exponential growth becomes correct. The first follows directly from the construction. The second can fail because small residual errors are amplified by an unstable feedback loop in the recurrence. The Jacobian of that loop, evaluated at the *true* generating function, is the quantity that decides the limiting obstruction.

This article develops that observation into a precise theorem, a sharp asymptotic law, and a small exact certificate applicable to a concrete polyomino recurrence. The contribution is not a further decimal improvement to the existing upper bound; it explains both the power and the ultimate limitation of continuing that particular improvement strategy.

13.2 The repository problem being continued

The inspected ProveIt snapshot contains the report *Finite-Prefix Corrections and Dual Certificates for Polyomino Growth* [[16](#)]. It builds on the formal project `Combinatorics/Polyominoes/KlarnerConstant`. The report's Section 8 proves eventual certificate existence when

$$\text{spr}(\Phi_X(t, A(t))) < 1,$$

and an obstruction under strict supercriticality and additional finite-prefix conditions. It explicitly leaves the equality case unclassified. It also shows, in a scalar example, that prefix-majorant growth can tend to 3 while true growth is 1.

The same report lists locating the actual fixed-map stability barrier and obtaining certified critical asymptotics among its proposed research projects. These are the specific continuation points addressed here. The numerical bounds reported by that source have distinct statuses:

4.5235 is the endpoint of its cited Lean development, whereas 4.498 is the source report’s computer-assisted, nonformalized upper bound. We do not reprove either endpoint in this article [16, 17].

[write] The report meant here is Part I above, read at the pin 74f7f5bdb, where its `article.tex` is the blob now printed: its Section 8 is [section 8](#), the equality case is the sentence after the proof of [theorem 8.1](#), the scalar example is [theorem 8.2](#), and the two research projects are questions 4 and 9 of [section 10](#). The Lean-verified bound is $\lambda \leq 9047/2000$ ([section 1.5](#)); nothing in Part II changes it.

13.3 Results proved here

The main results are as follows. The theorem numbers give the precise hypotheses; in particular, irreducibility and nonlinearity must not be silently discarded.

Question	Answer	Location
Critical certificates	At an irreducible Perron value ≥ 1 , a certificate exists exactly when the omitted defect is zero.	Theorem 15.2
Limiting accuracy	The prefix radii tend to the first spectral crossing, or the true radius when there is no earlier crossing.	Theorem 16.3
Rate of approach	At an interior nonlinear crossing, the radius gap is asymptotic to an explicit constant times the square root of the omitted defect.	Theorem 18.2
Local branch shape	The scaled solution has a quadratic universal profile; its square-root singular amplitude has a quarter-power law.	Theorem 19.1
Concrete limitation	The fixed 17-variable polyomino map cannot yield prefix-majorant growth below 4.3149, whatever the prefix length.	Theorem 22.2

13.4 Relation to prior methods and novelty

Bui’s convolutional approach supplies the polyomino recurrence and a finite-parameter method for certifying its growth [1]. We checked version 2, revised May 6, 2026, rather than relying on the differently titled first version. The exact-prefix construction, its monotonicity, the strict stability criterion, and the scalar limit 3 are already in the repository report [16]. They are restated and proved below to make the extension self-contained, not presented as discoveries here.

Inductive upper bounds and Perron-directed certificate searches are also established techniques for positive polynomial systems. In particular, Winkler and Katoen characterize strict inductive upper bounds using the Jacobian at a least fixed point [20]; Newton methods for monotone polynomial systems have a substantial independent literature [21]. Square-root singularities of positive functional systems are classical [22]. Neither Perron–Frobenius theory nor the local fold mechanism is claimed as new.

The proposed contribution is their precise application to *a moving family of exact-prefix corrections*: the full-defect critical alternative, the exact spectral saturation theorem, the sharp normalization in terms of the omitted defect without a regularity assumption on its coefficient sequence, the accompanying amplitude law, and the explicit all-prefix polyomino obstruction. We did not find these statements in the inspected repository continuation or the primary sources reviewed. This is a bounded novelty assessment, not an exhaustive priority claim.

13.5 What is not being claimed

This article does not determine Klarner's constant, improve the source's 4.498 upper bound, establish a lower bound 4.3149 for the true constant, or resolve Bui's separate question about enlarging the geometric neighborhood system. It does not certify the repository's entire mathematical corpus. Its analytic results have ordinary written proofs; its numerical application has an exact arithmetic checker, but the marked enumeration and the new theorems are not kernel-checked here.

14 Positive systems and exact-prefix corrections

14.1 Standing definitions

For vectors, inequalities are componentwise. For power series, $F \leq G$ means that every coefficient of every component of $G - F$ is nonnegative. A series has zero constant term unless stated otherwise.

Fix a dimension $d \geq 1$ and

$$\Phi(\xi, X) \in \mathbb{R}_{\geq 0}[\xi, X_1, \dots, X_d]^d, \quad \Phi(0, 0) = 0.$$

We call the system *proper* when

$$L = \Phi_X(0, 0) \text{ is nilpotent.} \quad (25)$$

This condition allows acyclic zero-size linear dependencies; it is weaker than requiring every monomial to contain ξ . Throughout, let

$$A \in \xi \mathbb{R}_{\geq 0}[[\xi]]^d, \quad A \leq \Phi(\xi, A). \quad (26)$$

The *full defect* and its prefix and tail are

$$D = \Phi(\xi, A) - A \geq 0, \quad D_N = [\xi^{\leq N}]D, \quad \mathcal{R}_N = D - D_N. \quad (27)$$

The exact profile prefix is $P_N = [\xi^{\leq N}]A$. Since all substituted series have zero constant term,

$$D_N = [\xi^{\leq N}]\Phi(\xi, P_N) - P_N. \quad (28)$$

Thus D_N can be computed from the prefix alone; no infinite-series oracle is required to construct a finite correction.

Define the positive tail map

$$\begin{aligned} \Psi_N(\xi, Y) &= \Phi(\xi, P_N + Y) - [\xi^{\leq N}]\Phi(\xi, P_N) \\ &= \Phi(\xi, P_N + Y) - P_N - D_N. \end{aligned} \quad (29)$$

Although the second line uses subtraction, the polynomial Ψ_N has nonnegative coefficients. Indeed, its terms involving Y are positive terms of the polynomial expansion, and its constant-in- Y part is $[\xi^{>N}]\Phi(\xi, P_N)$.

14.2 Formal fixed points and comparison

Lemma 14.1 (Graded convergence). *A proper nonnegative polynomial map $T(\xi, Y)$ with $T(0, 0) = 0$ has a unique zero-constant formal fixed point Y . Iteration from any zero-constant formal vector converges coefficient by coefficient to Y , with every coefficient eventually constant. Iteration from zero is nondecreasing. If $S \geq 0$ has zero constant term and $S \leq T(\xi, S)$, then $S \leq Y$. If $S \geq T(\xi, S)$, then $S \geq Y$.*

Proof. Suppose two zero-constant vectors agree below degree m . The degree- m part of the difference of their images under T is L times the corresponding difference, where $L = T_Y(0, 0)$. All other contributions contain either a positive power of ξ or another positive-degree series factor, so they only involve lower-degree differences. If $L^q = 0$, q iterations remove the degree- m difference once lower degrees agree. Induction on m gives eventual agreement, uniqueness, and coefficientwise stabilization of iteration. The stabilized coefficients solve the equation.

Nonnegative coefficients make T monotone. Thus its iterates from zero increase. For a subsolution, $S \leq T^k(S)$ for every k , so passage to the stabilized coefficients gives $S \leq Y$. The reversed argument applies to a supersolution. This last argument is important: a formal subsolution need not lie below a particular *finite* iterate from zero. $\square$

[write] *Restatement.* Existence, uniqueness, stabilization and subsolution comparison are Part I's [lemma 3.2](#); the convergence from an arbitrary zero-constant start and the supersolution comparison are the manuscript's additions. The proof above, by degree-wise contraction rather than by inverting $I - L$, is kept as a second proof.

Proposition 14.2 (Prefix majorants; prior construction). *Let Y_N be the formal fixed point of Ψ_N and put $B_N = P_N + Y_N$. Then*

$$B_N = \Phi(\xi, B_N) - D_N, \quad A \leq B_{N+1} \leq B_N, \quad [\xi^{\leq N}]B_N = P_N. \quad (30)$$

In particular, $B_N \rightarrow A$ coefficientwise.

Proof. The linear part of Ψ_N at $(0, 0)$ is L , so Lemma 14.1 applies. Its forcing begins above degree N ; iteration from zero therefore shows that Y_N begins above degree N . Equation (30) follows from the definition.

The true tail $S = A - P_N$ satisfies

$$\Psi_N(\xi, S) = S + \mathcal{R}_N \geq S,$$

so $A \leq B_N$. Hence $B_N - P_{N+1} \geq 0$, and

$$\Psi_{N+1}(\xi, B_N - P_{N+1}) = B_N - P_{N+1} - (D_{N+1} - D_N) \leq B_N - P_{N+1}.$$

Formal supersolution comparison gives $B_{N+1} \leq B_N$. The prefix identity gives coefficientwise convergence. $\square$

[write] *Restatement.* This is Part I's [theorems 3.3](#) and [3.6](#), as the manuscript's title "prior construction" says; the monotonicity proof above, by supersolution comparison instead of degree induction, is kept as a second proof.

14.3 Numeric certificates and the relevant radius

For a nonnegative vector series F , define its *system radius* as the minimum of its component radii of convergence. This convention is useful without first proving that all radii coincide. Let R be the system radius of A and ρ_N that of B_N , and write

$$\mu_N = 1/\rho_N = \max_i \limsup_{n \rightarrow \infty} ([\xi^n]B_{N,i})^{1/n}.$$

The conventions $1/\infty = 0$ and $1/0 = \infty$ apply.

Definition 14.3 (Upper certificate). A real upper certificate at a positive evaluation t and level N is a finite vector v satisfying

$$v \geq P_N(t), \quad v \geq \Phi(t, v) - D_N(t). \quad (31)$$

The tail condition $v \geq P_N(t)$ is part of the definition.

Lemma 14.4 (Exact numeric meaning). *A certificate at (N, t) exists if and only if $B_N(t)$ is finite coordinatewise. Every such certificate dominates $B_N(t)$ and hence $A(t)$ whenever the latter is finite. When $B_N(t)$ is finite, it is itself a certificate. Existence at level N implies existence at every later level, at the same evaluation point.*

Proof. Write $y = v - P_N(t) \geq 0$. Condition (31) is $\Psi_N(t, y) \leq y$. Numeric iteration from zero stays below y and increases. Evaluation commutes with the nonnegative coefficientwise limit: this follows by interchanging two increasing suprema, over iteration number and coefficient cutoff. Therefore the limit is the finite sum $Y_N(t)$. Polynomial continuity gives its fixed-point identity. Conversely a finite sum $Y_N(t)$ satisfies that identity by evaluation, and so supplies a certificate. The domination of A and persistence at later levels follow from Proposition 14.2. $\square$

[write] *Restatement.* The “if” direction with its domination and the persistence are Part I’s theorems 3.4 and 3.6 (whose certificate conditions (12) are (31), with ζ written t); the converse, that a finite $B_N(t)$ is itself a certificate, is not stated in Part I.

A certificate consequently yields $[\xi^n]B_{N,i} \leq v_i t^{-n}$. Failure of a certificate at t implies $\rho_N \leq t$, but does not by itself imply the strict inequality $\rho_N < t$: a series can diverge exactly at its radius. This distinction will matter in the linear example of Section 21.

15 The critical spectral case: an exact alternative

Fix $t > 0$ for which $A(t)$ is finite, and abbreviate

$$a = A(t), \quad J = \Phi_X(t, a).$$

A nonnegative matrix is called irreducible when its directed dependency graph is strongly connected. We use the Perron–Frobenius theorem in its standard finite-dimensional form: an irreducible nonnegative matrix has strictly positive left and right eigenvectors for its spectral radius, and that eigenvalue is algebraically simple [20].

Lemma 15.1 (The full-defect inequality). *Every certificate v gives $h = v - a \geq 0$ satisfying*

$$h \geq Jh + Q_t(h) + \mathcal{R}_N(t), \quad Q_t(h) = \Phi(t, a + h) - \Phi(t, a) - Jh \geq 0. \quad (32)$$

If $v = B_N(t)$, equality holds.

Proof. Lemma 14.4 gives $v \geq a$. Also $\Phi(t, a) = a + D(t)$. Subtracting a from the certificate inequality and expanding at a gives (32). The remainder is nonnegative because the polynomial has nonnegative coefficients and $a, h \geq 0$. $\square$

Theorem 15.2 (Critical and supercritical alternative). *Suppose J is irreducible and $\text{spr}(J) \geq 1$. For each N , the following are equivalent:*

- (i) *a real upper certificate exists at (N, t) ;*
- (ii) $\mathcal{R}_N(t) = 0$;
- (iii) $D = D_N$ as formal series;
- (iv) $B_N = A$ as formal series.

Thus an infinite full defect rules out every finite-prefix certificate at an irreducible critical evaluation, not merely at a strictly supercritical one.

Proof. Let $\ell > 0$ be a left Perron vector, with eigenvalue $\Lambda \geq 1$. Multiplying (32) by ℓ^T gives

$$(1 - \Lambda)\ell^T h \geq \ell^T Q_t(h) + \ell^T \mathcal{R}_N(t). \quad (33)$$

The left side is nonpositive and both terms on the right are nonnegative. Because $\ell > 0$, a nonzero defect tail makes the right side strictly positive. Hence (i) implies (ii).

All coefficients of $\mathcal{R}_N$ are nonnegative and $t > 0$, so its value vanishes exactly when every coefficient vanishes. This proves (ii) $\Leftrightarrow$ (iii). When $D = D_N$, the series $A - P_N$ is a fixed point of Ψ_N . Formal uniqueness gives $B_N = A$, proving (iii) $\Rightarrow$ (iv). Conversely, the equations for A and B_N give $D = D_N$ if $A = B_N$. Finally, (iv) and finiteness of $A(t)$ give (i) by Lemma 14.4. $\square$

Corollary 15.3 (Uniqueness at a nonlinear critical point). *Under Theorem 15.2, assume also $a > 0$ and that $\Phi(t, X)$ contains a monomial of total X -degree at least two. Whenever a certificate exists, the only one is $v = a$. In particular a rational certificate at this critical or supercritical evaluation exists exactly when $D = D_N$ and $a \in \mathbb{Q}^d$.*

Proof. If $h \neq 0$ in (32), then $h \geq Jh$ and irreducibility imply $h > 0$: a positive coordinate propagates along paths of powers of J . The stated nonlinear monomial then makes $Q_t(h)$ nonzero. Thus $\ell^T Q_t(h) > 0$, contradicting (33) even when $\mathcal{R}_N = 0$. Consequently $h = 0$. $\square$

Remark 15.4 (Why this is stronger than a prefix-Jacobian test). The repository's earlier obstruction uses $J_N = \Phi_X(t, P_N(t))$. At a true critical point, every finite J_N can still be strictly subcritical. The key improvement is to first establish $v \geq A(t)$ and expand at the *full* profile. This replaces the finite-prefix forcing by the uncorrected full defect. No limit interchange at a singularity is needed.

16 Exact spectral saturation of the hierarchy

16.1 Subcritical existence with explicit slack

For $0 < t < R$, set $a(t) = A(t)$ and $J(t) = \Phi_X(t, a(t))$.

Proposition 16.1 (Quantitative supersolution test). *Suppose $w > 0$, $\alpha > 0$, and $C \geq 0$ satisfy*

$$(I - J(t))w \geq \alpha \mathbf{1},$$

and, for $0 \leq s \leq s_0$,

$$\Phi(t, a + sw) - \Phi(t, a) - sJ(t)w \leq Cs^2 \mathbf{1}.$$

Let $r_N = \|\mathcal{R}_N(t)\|_\infty$. If

$$0 < s \leq s_0, \quad r_N + Cs^2 < \alpha s, \quad (34)$$

then $v = a + sw$ is a strict upper certificate. If Φ, D_N, t have rational data, a sufficiently close rational vector is also a certificate.

Proof. The right side of the corrected equation at $a + sw$, minus $a + sw$, is

$$\mathcal{R}_N(t) - s(I - J(t))w + Q_t(sw) \leq (r_N - \alpha s + Cs^2) \mathbf{1} < 0.$$

Also $a + sw > P_N(t)$, so both inequalities have strict slack. Polynomial continuity preserves them in a small neighborhood, which contains rational vectors. $\square$

For $C > 0$, the quadratic budget has maximum $\alpha^2/(4C)$, reached at $s = \alpha/(2C)$ when this value is allowed by s_0 . This gives an explicit reason to compare the remaining defect with the *square* of the spectral gap. The test is analytic as stated: $a(t)$ need not be known exactly. A directly computable version replaces $a, J(t), \mathcal{R}_N(t)$ by $P_N(t), \Phi_X(t, P_N(t)), b_N(t)$, where

$$b_N(t) = ([\xi^{>N}] \Phi(\xi, P_N(\xi)))(t). \quad (35)$$

The same proof then constructs $P_N(t) + sw$ using finite polynomial data.

Corollary 16.2 (Eventual subcritical certificates; prior strict criterion). *If $0 < t < R$ and $\text{spr}(J(t)) < 1$, all sufficiently large N admit a strict certificate at t . If Φ and t are rational, and the profile coefficients are rational, the certificate may be chosen rational.*

Proof. Take $w = (I - J(t))^{-1} \mathbf{1} = \sum_{k \geq 0} J(t)^k \mathbf{1} > 0$ and $\alpha = 1$. A polynomial Taylor remainder provides a finite C for $0 \leq s \leq 1$. Choose a fixed small $s > 0$ with $Cs^2 < s/2$. Since the nonnegative series D converges at t , its tail r_N tends to zero. For large N , $r_N < s/2$, giving the strict inequality (34). Rationality follows from strict slack. $\square$

[write] *Restatement.* This is Part I's [theorem 8.1\(a\)](#), as the manuscript's title "prior strict criterion" says. The proof above, through [Proposition 16.1](#) and expansion at the true profile, is kept as a second proof; Part I expands at the prefix profile.

16.2 The exact limiting radius

Theorem 16.3 (Spectral saturation). *Assume $R > 0$, D is not a polynomial, and $J(t)$ is irreducible for $0 < t < R$. Define*

$$r_* = \sup\{t \in (0, R) : \text{spr}(J(t)) < 1\}, \quad (36)$$

with $r_ = R$ if the spectral radius remains below one throughout $(0, R)$. Then*

$$\rho_N \uparrow r_*, \quad \mu_N \downarrow 1/r_*. \quad (37)$$

If $r_ < R$, then $\text{spr}(J(r_*)) = 1$ and no finite-prefix series converges coordinatewise at r_* . In particular, exact coefficient stabilization does not force the limiting growth to equal the growth of A .*

Proof. All entries of $J(t)$ are nonnegative power series, so $J(t)$ and its spectral radius are nondecreasing in t . At zero the Jacobian is nilpotent, hence its spectral radius is zero. Continuity shows that the subcritical set is a nonempty initial interval. If its supremum is interior to $(0, R)$, continuity also gives the value one there.

For every $t < r_*$, [Corollary 16.2](#) gives a certificate for large N . Thus $\liminf \rho_N \geq t$, and then $\liminf \rho_N \geq r_*$. [Proposition 14.2](#) makes the radii nondecreasing and bounds them by R . If $r_* = R$, this proves the claim. If $r_* < R$, the full defect has a nonzero tail at every level, so [Theorem 15.2](#) and [Lemma 14.4](#) give $\rho_N \leq r_*$ and nonconvergence at r_* . This proves the result in the remaining case. $\square$

Remark 16.4 (The finite-defect exception is exact). If D is a polynomial, then $B_N = A$ for all sufficiently large N , regardless of the Jacobian farther along the real axis. This is not a small technical exception to [Theorem 16.3](#); it is a genuinely different regime with finite termination.

17 Tail flatness without coefficient regularity

A sharp rate theorem must control how the omitted defect changes when the evaluation point moves with N . A naive argument might assume that its coefficients have a smooth asymptotic form or are not lacunary. Neither assumption is needed.

Lemma 17.1 (A square-root tail estimate). *Let $H(\xi) = \sum_{n \geq 0} h_n \xi^n$ have nonnegative coefficients and radius of convergence greater than $r > 0$. Assume H is not a polynomial, and write*

$$H_N(\xi) = \sum_{n > N} h_n \xi^n, \quad \varepsilon_N = \sqrt{H_N(r)}.$$

For every fixed $K < \infty$, uniformly on $|t - r| \leq K\varepsilon_N$,

$$\frac{H_N(t)}{H_N(r)} \longrightarrow 1, \quad H'_N(t) = o(\varepsilon_N), \quad H''_N(t) = o(1). \quad (38)$$

Proof. Choose $0 < a < r < b$ inside the convergence disk. For $t \in [a, b]$, Cauchy–Schwarz gives

$$\begin{aligned} H'_N(t) &= t^{-1} \sum_{n > N} n h_n t^n \\ &\leq a^{-1} \sqrt{H_N(t)} \sqrt{\sum_{n > N} n^2 h_n b^n}. \end{aligned} \quad (39)$$

The final sum tends to zero because the power series is analytic beyond b . Every tail is strictly positive on $(0, \infty)$, so

$$\left| \frac{d}{dt} \sqrt{H_N(t)} \right| \leq \frac{1}{2a} \sqrt{\sum_{n > N} n^2 h_n b^n} = c_N \longrightarrow 0.$$

For $|t - r| \leq K\varepsilon_N$, it follows that

$$\left| \frac{\sqrt{H_N(t)}}{\varepsilon_N} - 1 \right| \leq Kc_N \longrightarrow 0.$$

This proves the first assertion. Substitution in (39) gives the second. Finally,

$$0 \leq H''_N(t) \leq a^{-2} \sum_{n > N} n(n-1) h_n b^n \longrightarrow 0.$$

All estimates are uniform on the specified shrinking interval. □

Corollary 17.2 (Vector version). *Let D be a nonnegative vector series analytic beyond r , let $\ell > 0$, and set*

$$\eta_N = \ell^T \mathcal{R}_N(r) > 0, \quad \varepsilon_N = \sqrt{\eta_N}.$$

Uniformly for $|t - r| \leq K\varepsilon_N$,

$$\ell^T \mathcal{R}_N(t) = \eta_N(1 + o(1)), \quad \|\mathcal{R}_N(t)\| = O(\eta_N), \quad \|\mathcal{R}'_N(t)\| = o(\varepsilon_N), \quad \|\mathcal{R}''_N(t)\| = o(1).$$

Proof. Apply Lemma 17.1 to $H = \ell^T D$. Each component and its nonnegative derivatives are bounded by a fixed multiple of the corresponding weighted scalar sum. Norm equivalence in finite dimension finishes the argument. □

The vector directions $\mathcal{R}_N(r)/\eta_N$ need not converge. The proof below only uses their boundedness and their fixed Perron projection. Thus oscillating component contributions and arbitrarily long coefficient gaps are allowed.

18 Sharp convergence at an interior nonlinear crossing

18.1 Hypotheses and invariant constants

Assume the hypotheses of Theorem 16.3, and suppose $r_* < R$. Assume also $a_* = A(r_*) > 0$ and that Φ contains a monomial of total X -degree at least two. Put

$$J_* = \Phi_X(r_*, a_*), \quad J_* u = u, \quad \ell^T J_* = \ell^T, \quad u, \ell > 0, \quad \ell^T u = 1.$$

Define

$$\kappa = \ell^T J'(r_*)u, \quad \beta = \frac{1}{2} \ell^T \Phi_{XX}(r_*, a_*)[u, u], \quad \eta_N = \ell^T \mathcal{R}_N(r_*). \quad (40)$$

The derivative J' is the *total* derivative along the true profile:

$$J'(t) = \Phi_{\xi X}(t, A(t)) + \Phi_{XX}(t, A(t))[A'(t), \cdot]. \quad (41)$$

For a vector-valued Hessian, the bracket means componentwise evaluation of its bilinear form.

Lemma 18.1 (Nondegeneracy is automatic here). *Under these assumptions, $\kappa > 0$, $\beta > 0$, and $\eta_N > 0$ for every N . If $\Lambda(t) = \text{spr}(J(t))$ near r_* , then*

$$1 - \Lambda(t) = \kappa(r_* - t) + O((r_* - t)^2). \quad (42)$$

Proof. The nonnegative analytic matrix $J(t)$ has nonnegative derivative. Some entry is non-constant: otherwise J_* would equal the nilpotent $J(0)$, contradicting its Perron value one. A nonconstant power series with nonnegative coefficients has strictly positive derivative at every positive point of convergence. Since $u, \ell > 0$, this gives $\kappa > 0$.

A nonlinear monomial evaluated at positive r_*, a_*, u contributes positively to at least one component of the Hessian evaluation; all other contributions are nonnegative. Hence $\beta > 0$. The infinite nonnegative defect gives $\eta_N > 0$. The simple Perron eigenvalue varies analytically near r_* , and differentiation of its eigenvalue equation gives $\Lambda'(r_*) = \ell^T J'(r_*)u = \kappa$. $\square$

Rescaling u to cu and ℓ to ℓ/c changes β to $c\beta$ and η_N to η_N/c . All the formulas below are invariant under this change. Thus the normalization $\ell^T u = 1$ leaves no ambiguity in any observable prediction.

18.2 Statement of the sharp law

Theorem 18.2 (Square-root saturation law). *Under the assumptions just stated,*

$$r_* - \rho_N \sim \frac{2\sqrt{\beta}}{\kappa} \sqrt{\eta_N}. \quad (43)$$

For all sufficiently large N , $\rho_N < r_$, $B_N(\rho_N)$ is finite, and*

$$B_N(\rho_N) - A(\rho_N) = u \sqrt{\eta_N/\beta} + o(\sqrt{\eta_N}). \quad (44)$$

The vector error is in any fixed finite-dimensional norm. Consequently

$$\mu_N - \frac{1}{r_*} \sim \frac{2\sqrt{\beta}}{\kappa r_*^2} \sqrt{\eta_N}. \quad (45)$$

The proof has two parts. A local reduction identifies the fold. A separate localization argument proves that it is the actual convergence boundary, not a spurious branch of the polynomial equations. Omitting the second part would leave a genuine gap.

18.3 Uniform localization of every possible certificate

Lemma 18.3 (No remote fixed point can evade the fold). *There exist constants $C, \delta_0 > 0$, independent of N , such that, for $r_* - \delta_0 < t < r_*$, every certificate at (N, t) implies*

$$\|B_N(t) - A(t)\| \leq C(r_* - t). \quad (46)$$

Proof. Let $h = B_N(t) - A(t) \geq 0$. The full-defect equality gives

$$h = J(t)h + Q_t(h) + \mathcal{R}_N(t), \quad (47)$$

so $h \geq J(t)h$. Irreducibility persists near r_* . The matrix

$$K(t) = I + J(t) + \cdots + J(t)^{d-1}$$

has every entry bounded below by a positive constant on a sufficiently small closed neighborhood of r_* . Since $K(t)h \leq dh$, there is $c > 0$ such that

$$h_i \geq c \|h\|_1 \quad (1 \leq i \leq d) \quad (48)$$

for every such h . This argument also covers $d = 1$.

Let $\ell(t) > 0$ be a continuous normalized left Perron vector. Its coordinates are uniformly bounded above and below. The positive nonlinear monomial in Φ , together with positivity of $A(t)$, ensures that at least one coefficient in the quadratic part of Q_t is uniformly positive. Using (48), we obtain

$$\ell(t)^T Q_t(h) \geq c_1 \|h\|_1^2$$

after decreasing the neighborhood if needed. This bound holds for all $h \geq 0$ satisfying (48), not just for small h : higher-degree terms of the expansion are nonnegative.

Multiplying (47) by $\ell(t)^T$ now gives

$$(1 - \Lambda(t))\ell(t)^T h \geq c_1 \|h\|_1^2.$$

Since $\ell(t)^T h \leq c_2 \|h\|_1$ and $1 - \Lambda(t) = O(r_* - t)$, division when $h \neq 0$ proves the bound. If $h = 0$, the bound is immediate (and a nonzero residual would in fact rule that case out). $\square$

18.4 Reduction to a scalar equation

Set $\delta = r_* - t$ and write

$$h = su + w, \quad \ell^T w = 0, \quad \Pi = I - u\ell^T.$$

On the subspace $\ell^T w = 0$, the map $I - J_*$ is invertible. Indeed, J_* preserves this subspace and its only eigenvector with eigenvalue one is a multiple of u . Other peripheral eigenvalues, including -1 , are harmless; primitivity is not required.

For an independent vector parameter r , consider

$$G(t, h, r) = h - \Phi(t, A(t) + h) + \Phi(t, A(t)) - r. \quad (49)$$

The analytic implicit function theorem applied to $\Pi G = 0$ gives

$$w = W(\delta, s, r), \quad W(0, 0, 0) = 0,$$

with

$$\|W(\delta, s, r)\| = O(|\delta s| + s^2 + \|r\|). \quad (50)$$

To see the absence of a term proportional to δ alone, note that $G(t, 0, 0) = 0$ identically. The absence of a term linear in s at $\delta = 0$ follows from $(I - J_*)u = 0$.

Substitute $r = \mathcal{R}_N(r_* - \delta)$ and define the scalar reduced equation

$$g_N(\delta, s) = \ell^T G(r_* - \delta, su + W(\delta, s, \mathcal{R}_N(r_* - \delta)), \mathcal{R}_N(r_* - \delta)). \quad (51)$$

The local fixed points are exactly its zeros.

Put $\varepsilon_N = \sqrt{\eta_N}$. On bounded sets of K, S , set $\delta = \varepsilon_N K$ and $s = \varepsilon_N S$. Taylor expansion, (50), and Corollary 17.2 give

$$F_N(K, S) := \frac{g_N(\varepsilon_N K, \varepsilon_N S)}{\varepsilon_N^2} \longrightarrow F(K, S) := \kappa KS - \beta S^2 - 1 \quad (52)$$

in C^2 on every bounded rectangle.

Here are the terms responsible for the limit. The projected linear part is

$$\ell^T (I - J(r_* - \delta))su = \kappa \delta s + O(\delta^2 |s|).$$

The leading nonlinear part is $-\beta s^2$. Contributions involving W are $O(\varepsilon_N^3)$ in this scaling, because $W = O(\varepsilon_N^2)$ and $\ell^T (I - J_*)W = 0$. Finally the projected forcing is $-\eta_N(1 + o(1))$. For the first and second scaled derivatives, Corollary 17.2 gives exactly the needed bounds $\mathcal{R}'_N = o(\varepsilon_N)$ and $\mathcal{R}''_N = o(1)$. This verifies the C^2 assertion, including the moving forcing; a fixed-parameter Taylor expansion alone would not suffice.

18.5 The fold and its identification with the true radius

The limiting equations $F = 0$ and $F_S = 0$ have the unique solution with $K, S > 0$

$$K_c = \frac{2\sqrt{\beta}}{\kappa}, \quad S_c = \frac{1}{\sqrt{\beta}}. \quad (53)$$

At this point the Jacobian of (F, F_S) with respect to (K, S) is

$$\begin{pmatrix} \kappa/\sqrt{\beta} & 0 \\ \kappa & -2\beta \end{pmatrix},$$

which is invertible. The C^2 convergence in (52) and the implicit function theorem give a nearby fold $(K_N, S_N) \rightarrow (K_c, S_c)$ for every sufficiently large N . At that fold,

$$(g_N)_\delta = \kappa \varepsilon_N / \sqrt{\beta} + o(\varepsilon_N) > 0, \quad (g_N)_{ss} = -2\beta + o(1) < 0. \quad (54)$$

The associated $h = \varepsilon_N S_N u + O(\varepsilon_N^2)$ is strictly positive and solves the corrected equation, hence gives a certificate.

We must show that this fold is the boundary of all certificates. Fix a small $a > 0$. At $K = K_c + a$, the limiting polynomial $F(K, S)$ has two positive simple roots. The implicit function theorem gives nearby roots of F_N , and their vectors h are positive; thus a certificate exists. At $K = K_c - a > 0$, no limiting positive root exists. If certificates existed there along a subsequence, Lemma 18.3 would make their scaled h bounded. The complementary equation forces $w = O(\varepsilon_N^2)$; a convergent subsequence of S would then solve $F(K_c - a, S) = 0$, a contradiction.

The convergence boundary is therefore trapped in an arbitrarily small fixed scaled neighborhood of K_c . In that neighborhood every possible fixed point is localized by Lemma 18.3, and every localized solution lies near S_c . The reduced function is strictly concave in s there

by (54); its unique local maximum crosses zero at (K_N, S_N) . Its derivative with respect to K at this crossing is positive. Consequently local roots exist on the larger- K side and none on the smaller- K side. No other roots can supply a certificate by the localization just proved. Thus

$$\rho_N = r_* - \varepsilon_N K_N.$$

At the fold the fixed point is finite. Lemma 14.4 identifies it with $B_N(\rho_N)$, since no other localized nonnegative fixed point exists there. Its height is $\varepsilon_N S_N u + O(\varepsilon_N^2)$. Together with (53), these facts prove (43) and (44). The reciprocal expansion proves (45), completing the proof of Theorem 18.2.

19 Universal branch shape and a quarter-power amplitude

Theorem 19.1 (Scaled branch and local singular amplitude). *Under Theorem 18.2, put*

$$S_-(K) = \frac{\kappa K - \sqrt{\kappa^2 K^2 - 4\beta}}{2\beta} \quad \left(K > \frac{2\sqrt{\beta}}{\kappa} \right).$$

Uniformly on compact subsets of this interval,

$$\frac{B_N(r_* - K\varepsilon_N) - A(r_* - K\varepsilon_N)}{\varepsilon_N} \longrightarrow u S_-(K). \quad (55)$$

For each sufficiently large fixed N , the minimal branch has a local expansion as real $t \uparrow \rho_N$,

$$B_N(t) = B_N(\rho_N) - a_N \sqrt{1 - t/\rho_N} + O(1 - t/\rho_N), \quad a_N > 0, \quad (56)$$

and

$$\boxed{a_N = u \sqrt{r_* \kappa} \beta^{-3/4} \eta_N^{1/4} + o(\eta_N^{1/4})}. \quad (57)$$

Proof. For fixed $K > K_c$, the two roots of the limiting reduced equation are simple and positive. The corresponding local solution vectors are ordered componentwise: their derivative with respect to s is $u + O(\varepsilon_N) > 0$. The minimal fixed point selects the smaller root. Localization excludes any other candidate, proving (55); compactness gives uniformity away from the fold.

For fixed N , the reduction is analytic in a neighborhood of its fold. Since $(g_N)_\delta > 0$ and $(g_N)_{ss} < 0$, the analytic implicit function theorem applied with the square-root parameter gives

$$s = s_N - \sqrt{\frac{2(g_N)_\delta}{-(g_N)_{ss}}} \sqrt{\rho_N - t} + O(\rho_N - t).$$

The sign is negative on the minimal branch. The vector derivative with respect to s is $u + O(\varepsilon_N)$; all other changes are analytic in $\rho_N - t$. Therefore

$$a_N = (u + O(\varepsilon_N)) \sqrt{\frac{2\rho_N(g_N)_\delta}{-(g_N)_{ss}}}.$$

Substitution of (54), $\rho_N \rightarrow r_*$, and $\varepsilon_N = \eta_N^{1/2}$ proves (57). $\square$

The local singularity is a genuine singularity in every component because $a_N > 0$. The theorem deliberately does not claim an unqualified global coefficient asymptotic $c_N \rho_N^{-n} n^{-3/2}$. Such a statement requires control of other singularities on the circle of convergence and the appropriate analytic continuation or periodicity hypotheses. A real positive-axis expansion alone does not establish those conditions.

19.1 Consequences for data requirements

If a particular system has

$$\eta_N \sim CN^p \theta^N, \quad 0 < \theta < 1,$$

then Theorem 18.2 gives

$$r_* - \rho_N \sim \frac{2\sqrt{\beta C}}{\kappa} N^{p/2} \theta^{N/2}.$$

The exponential rate is halved. To make the radius gap of size ϵ , the leading required prefix length is

$$N = \frac{2 \log(1/\epsilon)}{|\log \theta|} + O(\log \log(1/\epsilon)),$$

when the displayed tail equivalent holds. The quarter-power amplitude also quantifies how the singularity becomes weaker while its location approaches the barrier. These are conditional consequences of a tail law, not assumptions used to prove the general theorem.

20 An exact scalar model: limit, rate, and certificates

20.1 The previously known limit

Take

$$A(\xi) = \frac{\xi}{1-\xi}, \quad \Phi(\xi, X) = \xi + X^2. \quad (58)$$

The inequality $A \leq \xi + A^2$ holds because the coefficient of ξ^n in A^2 is $n-1$ for $n \geq 2$. The defect is

$$D(\xi) = \frac{\xi^3}{(1-\xi)^2} = \sum_{n \geq 3} (n-2) \xi^n. \quad (59)$$

The true radius is 1, but

$$J(t) = \frac{2t}{1-t}$$

first reaches one at $r_* = 1/3$. Thus the corrected radii tend to $1/3$ and their growth tends to 3. This limit and its interpretation were already established in the repository [16]. [write] It is Part I's theorem 8.2, whose proof is not repeated here; what follows is new in the manuscript.

For $N \geq 2$, the corrected equation and its selected root are

$$B_N = \xi + B_N^2 - D_N(\xi), \quad B_N(\xi) = \frac{1 - \sqrt{1 - 4\xi + 4D_N(\xi)}}{2}. \quad (60)$$

The branch is the one with zero constant term. The positive-tail construction, rather than the signed appearance of the radicand, guarantees nonnegative coefficients.

20.2 A sharp equivalent and an exact root equation

Here $u = \ell = 1$, $\beta = 1$, and

$$\kappa = J'(1/3) = \frac{9}{2}.$$

A direct geometric-series computation gives

$$\mathcal{R}_N(t) = \frac{t^{N+1}((N-1) - (N-2)t)}{(1-t)^2}, \quad \eta_N = \mathcal{R}_N(1/3) = \frac{2N-1}{4 \cdot 3^N}. \quad (61)$$

Therefore the new sharp equivalents specialize to

$$\boxed{\frac{1}{3} - \rho_N \sim \frac{2}{9} \sqrt{2N-1} 3^{-N/2}, \quad \mu_N - 3 \sim 2 \sqrt{2N-1} 3^{-N/2}.} \quad (62)$$

The singular amplitude satisfies

$$a_N \sim \sqrt{\frac{3}{2}} \left(\frac{2N-1}{4 \cdot 3^N} \right)^{1/4}. \quad (63)$$

These constants can also be checked without the general theorem. Since

$$1 - 4t + 4D(t) = \frac{(1-3t)^2}{(1-t)^2},$$

the exact root equation for ρ_N is

$$(1 - 3\rho_N)^2 = 4\rho_N^{N+1}((N-1) - (N-2)\rho_N). \quad (64)$$

For $N \geq 3$ there is exactly one root in $(1/4, 1/3)$. Indeed, $c_N(t) = t - D_N(t)$ is strictly increasing on $[1/4, 1/3]$: $D'_N(t) < D'(t) \leq 1$ there. Also $c_N(1/4) < 1/4$ and $c_N(1/3) > 1/4$. The root is where $c_N = 1/4$. At that point $v = 1/2 > P_N(\rho_N)$ is a valid certificate. Immediately to its right the quadratic has no real root, proving that this is exactly the radius.

To extract the equivalent directly, first the root equation gives $\rho_N \rightarrow 1/3$. Its positive square root gives

$$\frac{1}{3} - \rho_N = \frac{2}{3} \rho_N^{(N+1)/2} \sqrt{(N-1) - (N-2)\rho_N}.$$

The right side is $O(\sqrt{N} 3^{-N/2})$, so $N(1/3 - \rho_N) \rightarrow 0$. It follows that $(3\rho_N)^{(N+1)/2} \rightarrow 1$. Substitution now yields exactly the first formula in (62); the second follows by reciprocation. This is an independent derivation of the leading constants.

20.3 Exact computation and numerical illustration

The supplied checker brackets the root of (64) by rational bisection, using exact integer arithmetic in every sign test. It performs 260 bisections for each displayed N . Both rational endpoints and their exact width are saved in `data/02-spectral-saturation-verification.json`. The entries below are rounded displays of those brackets; the last column divides the gap by the proposed leading term in (62).

N	ρ_N (rounded)	$\rho_* - \rho_N$	ratio to leading term
4	0.287371536944	4.59618×10^{-2}	0.703563009
8	0.323938104014	9.39523×10^{-3}	0.884217871
16	0.333145629046	1.87704×10^{-4}	0.995350175
32	0.33333292359	4.09748×10^{-8}	0.999998001
64	0.333333333333	1.35148×10^{-15}	1.000000000
128	0.333333333333	1.03347×10^{-30}	1.000000000

Table 9: The scalar radius approaches its barrier at the predicted half-rate. The exact brackets, not the rounded decimal display, are the certificates.

For $N = 64$ and 128 , the rounded radius visually equals $1/3$, but the positive gap is separately displayed. No equality with $1/3$ is being asserted. The asymptotic theorem is proved analytically; agreement at six finite values is a consistency check, not a proof for all N .

21 Examples that delimit the hypotheses

21.1 A supercritical system that becomes exact in finitely many steps

Let $A(\xi) = \xi$ and $\Phi(\xi, X) = \xi + X^2$. Then $D = \xi^2$, so $B_N = A$ for $N \geq 2$. Nevertheless $\Phi_X(t, A(t)) = 2t$ is critical at $t = 1/2$ and supercritical afterward. The positive tail map is

$$\Psi_N(t, y) = 2ty + y^2 \quad (N \geq 2),$$

whose least fixed point is zero even when its derivative exceeds one. This is precisely the zero-forcing exception in Theorem 15.2. It disproves any formulation that rules out all supercritical certificates without examining the residual defect.

21.2 Irreducibility cannot be removed

Consider the independent two-component system

$$\Phi_1(\xi, X) = \xi + X_1^2, \quad \Phi_2(\xi, X) = \xi + \xi X_2,$$

with

$$A_1(\xi) = \xi, \quad A_2(\xi) = \frac{\xi}{1 - \xi/2}.$$

The defects are

$$D_1 = \xi^2, \quad D_2 = \frac{\xi^2}{2(1 - \xi/2)}.$$

At $t = 1/2$ the true-profile Jacobian is $\text{diag}(1, 1/2)$ and has spectral radius one. For every $N \geq 2$, the first component is already exact, while the second corrected series is finite because its linear denominator is $1 - \xi$. Thus certificates exist although the total defect tail is nonzero. The critical block receives no residual forcing. A reducible extension must track which blocks are forced and how they feed one another.

21.3 Linearity gives a different convergence regime

The scalar second component of the preceding example has true radius 2 and Jacobian $J(t) = t$, hence spectral barrier 1. Its corrected equation is

$$B_N = \xi + \xi B_N - D_N(\xi).$$

Subtracting the true equation gives the exact identity

$$B_N(\xi) - A(\xi) = \frac{\mathcal{R}_N(\xi)}{1 - \xi}.$$

Every finite level has a nonzero positive residual at $\xi = 1$, so every radius is *exactly* 1 and the boundary value diverges. There is no nonzero square-root radius gap. Here $\beta = 0$; the nonlinear hypothesis of Theorem 18.2 is essential.

21.4 Primitivity is not required

Let $A_1 = A_2 = \xi/(1 - \xi)$ and

$$\Phi_1 = \xi + X_2^2, \quad \Phi_2 = \xi + X_1^2.$$

The minimal branch remains symmetric and equals the scalar example in both coordinates. At $t = 1/3$ the Jacobian is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with eigenvalues 1 and -1 . The complement of the Perron direction remains invertible for $I - J$. Thus the sharp law applies despite period two in the dependency matrix. This matrix issue is separate from possible periodicity of generating-function coefficients.

22 An exact all-prefix barrier for the polyomino map

22.1 The finite input and the infinite conclusion

The repository report supplies a 17-component marked profile and a 53-monomial positive map, transcribed in Section 26. The variables are ordered

$$(c, d, e, f, g, h, p, q, r, s, t, u, v, w, x, y, z).$$

In this section and the appendix, the *size variable* is ξ , so it is not confused with the last coordinate z . [write] (In Part II as printed, ξ is the size variable throughout; see [section 12.2](#). The profile is Part I's [section A](#) and data/profiles_18.csv; the map is (6).) The source data list all marked coefficients through size 18 [19, 18]. We use those values as input. The checker replays their partition and recurrence consistency identities, but does not independently enumerate size-18 polyominoes. [write] (Part I's intake regenerated that table through size 18 from the final enumerator source, and through size 10 with an independent enumerator; see [section D](#). It is still not kernel-checked.)

A finite lower bound on the Jacobian already suffices to rule out *every* longer prefix. The following elementary lemma makes the certificate format explicit and avoids any reliance on a floating-point eigenvalue.

Lemma 22.1 (A rational spectral obstruction). *Let $J \geq 0$ be irreducible, let $b \geq 0$ be nonzero, and suppose there exist $w > 0$ and $\gamma > 1$ with $Jw \geq \gamma w$. There is no $y \geq 0$ satisfying $y \geq b + Jy$.*

Proof. Iteration gives $y \geq \sum_{k=0}^{d-1} J^k b$. Strong connectivity and $b \neq 0$ make this sum strictly positive, so $y > 0$. Put $m = \min_i y_i/w_i > 0$ and choose an index i attaining the minimum. Then

$$(Jy)_i \geq m(Jw)_i \geq m\gamma w_i > mw_i = y_i,$$

contradicting $y \geq Jy$. Every ingredient is a finite matrix or graph inequality. $\square$

22.2 The certified witness

Set

$$t_0 = \frac{10000}{43149}, \quad \gamma = \frac{1000001}{1000000}, \quad J_{18} = \Phi_X(t_0, P_{18}(t_0)). \quad (65)$$

The positive integer vector in Table 10 satisfies

$$\boxed{J_{18}w \geq \gamma w > w.} \quad (66)$$

The 17 inequalities are checked using exact rational arithmetic. A second implementation evaluates the directional derivative by dual-number multiplication and agrees exactly with the matrix-vector product. The smallest ratio $(J_{18}w)_i/w_i - 1$, rounded only for display, is 0.000004934886683313896, safely greater than the certified 0.000001.

Theorem 22.2 (A 4.3149 barrier for every prefix). *For the repository's fixed 17-variable map and its true marked profile, with the size-18 profile taken as the supplied enumerative input, no finite-prefix upper certificate exists at $t_0 = 10000/43149$. Every corrected system has growth at least*

$$\mu_N \geq \frac{43149}{10000} = 4.3149. \quad (67)$$

The same radius bound applies to its g component, which is the component used to majorize the unmarked fixed-polyomino counts.

coordinate	w_i
c	53,709,887,569
d	112,782,022,377
e	231,753,937,549
f	1,000,000,000,000
g	486,645,549,886
h	301,166,180,537
p	513,359,385,008
q	254,894,013,881
r	120,854,740,222
s	188,385,644,383
t	169,174,579,449
u	356,544,026,070
v	97,749,895,723
w	73,183,659,458
x	71,425,518,584
y	47,671,677,169
z	28,731,292,790

Table 10: An integer witness for (66). The column headed w_i is a vector entry, not the map coordinate named w .

Proof. For $M \geq 18$, nonnegativity of the true coefficients gives $P_M(t_0) \geq P_{18}(t_0)$, hence

$$J_M w \geq J_{18} w \geq \gamma w, \quad J_M = \Phi_X(t_0, P_M(t_0)).$$

All prefix coordinates are positive, and the map's dependency graph is strongly connected. The checker verifies this graph directly; positivity means its edges coincide with those of J_M .

The first row of the map is $\Phi_c = \xi + \xi e$. For every size M , the vertical column of M cells has an e -pattern occurrence at its bottom cell. In the source definition of e , the anchor is occupied while its horizontal neighbors and the three positions immediately below are absent; the vertical column has exactly this local configuration. Thus the true coefficient $e_M \geq 1$. It follows that the finite-prefix forcing

$$b_M(t_0) = ([\xi^{>M}] \Phi(\xi, P_M(\xi)))(t_0)$$

has a strictly positive c coordinate, containing $e_M t_0^{M+1}$.

A certificate would yield a tail $y \geq 0$ with $y \geq \Psi_M(t_0, y) \geq b_M(t_0) + J_M y$. Lemma 22.1 rules this out. Hence no level $M \geq 18$ has a certificate. An earlier certificate would persist to level 18 by Lemma 14.4, so every earlier level is excluded too. This proves the system-radius and growth assertions.

For completeness, when $M \geq 18$ the linear-in- Y part of the positive tail map is the formal matrix $\Phi_X(\xi, P_M(\xi))$. Each dependency edge contains a positive monomial $c \xi^k Y_j$. Therefore $Y_{M,i} \geq c \xi^k Y_{M,j}$ along every edge. Strong connectivity forces all nonzero tail components to have the same radius, and the positive forcing propagates to every component. In particular the g component has the system radius. For earlier levels, componentwise nesting with B_{18} gives the same upper bound on the g radius. $\square$

22.3 What this numerical result means

The theorem is an obstruction to an entire infinite hierarchy, obtained from one finite profile and one short rational witness. It shows that, even with arbitrarily long exact prefixes and arbitrarily sophisticated supersolution searches, the *unchanged* map cannot certify an upper growth constant below 4.3149 by this method.

It does not say that the true polyomino growth is at least 4.3149. The logical direction is

$$\text{true counts} \leq \text{corrected majorants},$$

so a lower bound for the majorants cannot be reversed into a lower bound for the true counts. Nor does it locate the actual saturation constant exactly: 4.3149 is only a certified lower bound on the limiting majorant growth. The true fixed-map barrier could be higher.

[write] *Tempting false readings.* (i) “ $\lambda \geq 4.3149$ ”: false as a reading of [theorem 22.2](#), which bounds the growth μ_N of the majorants B_N from below, while λ lies below every μ_N . (ii) “4.3149 < 4.498, so Part II improves Part I’s bound”: false; 4.3149 is not an upper bound on anything. It is a floor under every upper bound that this fixed map and exact prefixes can certify. Part I’s certificates 4.5233, . . . , 4.498 ([table 3](#)) and the Lean-verified 4.5235 all lie above it, as they must. The upper bounds on λ in ProveIt remain 4.498 (computer-assisted, [theorem 4.1](#)) and 9047/2000 (Lean-verified).

The calculation also does not assume that the full marked profile converges at t_0 . It uses only finite prefixes, a uniform nonzero-forcing construction, and certificate persistence. This makes it valid even before the hypotheses needed to apply the sharp interior-crossing theorem to the actual polyomino profile have been established.

22.4 Discovery and replay are separate

The optional script code/02-spectral-saturation-discover.py finds a numerical positive eigenvector, scales and rounds it to integers, and exports the witness. That procedure is not trusted by the proof. The replay script code/02-spectral-saturation-verify.py imports no numerical library and ignores all floating-point discovery fields. It recomputes the exact rational Jacobian from the supplied coefficient profile and the displayed monomial map.

The verifier additionally checks nilpotence of the zero-size linear part (the index is 3), the 53-monomial count, strong connectivity, all 323 component/degree defect checks through degree 18 including degree zero, and 95 checks for the five partition identities including degree zero. These are useful transcription checks, not a substitute for the combinatorial recurrence proof or the marked enumeration.

23 Reproducibility and proof audit

23.1 What is in the package

[write] The file names below are those shipped in ProveIt; the delivered package used the same names without the prefix 02-spectral-saturation-. The manuscript’s own article.tex, article.pdf and README.md are not shipped (Part II is their text), and neither is its copy of data/profiles_18.csv, which is byte-identical to Part I’s file of that name and is replaced by it. The two small TeX tables in data/ are included by the source. The other retained files are:

- code/02-spectral-saturation-bui_model.py, the transcribed map and rational Jacobian; code/02-spectral-saturation-verify.py, the exact checker; and optional code/02-spectral-saturation-discover.py, the numerical witness search.
- data/profiles_18.csv, the inherited marked profile (Part I’s file); data/02-spectral-saturation-bui_spectral_certificate.json, the integer witness; and data/02-spectral-saturation-verification.json and data/02-spectral-saturation-verification.txt, the recorded replay results, including exact scalar brackets. *Not to be confused with* Part I’s own data/verification.json and data/verification.txt, which record the 4.498 replay.

- `02-spectral-saturation-SOURCES.md`, `02-spectral-saturation-PROOF_STATUS.md`, and a build script `code/02-spectral-saturation-build.sh`.

To replay the checks without changing retained outputs, run

```
python3 code/verify.py
```

To regenerate the verification record and table inputs, run

```
python3 code/verify.py --write
```

[write] These two commands are the delivered ones and assume the delivered layout. In ProveIt they must not be run in place: the renamed scripts import `bui_model`, which is shipped as `code/02-spectral-saturation-bui_model.py`, and `-write` writes `data/verification.json`, which is Part I's record. The report's README gives a recipe that rebuilds the delivered layout in a scratch copy; there the checker's output equals the shipped record and tables. Only the optional discovery script needs NumPy. The exact checker uses Python's standard library, principally arbitrary-precision integers and `fractions.Fraction`. Failure of any required check raises an exception; it does not silently downgrade to a floating-point comparison.

23.2 Dependency structure of the mathematical claims

The critical theorem depends only on formal comparison, positivity of the full defect, and finite-dimensional Perron–Frobenius theory. The spectral limit additionally uses eventual subcritical supersolutions and convergence of nonnegative tails. The sharp law adds analyticity at an interior crossing, a nonlinear positive Hessian direction, the tail-flatness lemma, a finite-dimensional implicit-function reduction, and the global localization lemma.

The numerical all-prefix obstruction follows a different chain:

$$\begin{aligned}
 &\text{supplied marked prefix} + \text{displayed polynomial map} \\
 &\quad \implies \text{exact rational matrix witness} \\
 &\quad \implies \text{no certificate at every level } M \geq 18 \\
 &\quad \implies \text{no certificate at any level, by persistence.}
 \end{aligned}$$

The nonzero forcing at arbitrary size is proved by the vertical-column construction, not extrapolated from the finite data. The checker verifies arithmetic at size 18; the infinite-level conclusion is the accompanying mathematical argument.

23.3 Adversarial checks on the argument

Several potential failure modes have explicit safeguards. The tail budget in (31) is mandatory. The expansion at $A(t)$ is used only after proving that every certificate dominates $A(t)$. Nonzero forcing is distinguished from finite exact termination. Irreducibility cannot be replaced by an unsupported claim that every nonnegative Perron vector is strictly positive. The rate theorem uses the total derivative (41), not the partial derivative in the size variable alone. The local fold is identified with the actual radius using Lemma 18.3. Tail-flatness is proved rather than inferred from a guessed coefficient asymptotic. No global coefficient transfer principle is invoked from the real-axis singularity alone.

These safeguards reduce identifiable risks; they do not constitute independent refereeing or machine verification. In particular the analytic implicit-function and localization arguments should receive independent mathematical review before any publication-level novelty or correctness claim is treated as settled.

23.4 A feasible formalization sequence

A Lean continuation can be divided into three independent layers. The first layer formalizes the polynomial prefix identity, positivity of Ψ_N , formal comparison, and persistence of numeric supersolutions. It can reuse the existing recurrence and growth interfaces described in the repository report [16, 17].

The second layer formalizes Lemma 22.1 and checks the rational vector in Table 10. That lemma needs neither complex analysis nor numerical eigenvalues. Its remaining combinatorial input is a verified marked prefix, together with the simple vertical-column occurrence lemma. The checker included here is not such a formal enumerator.

The third layer develops the analytic saturation theorem and the sharp fold analysis. Its reusable targets include a theorem for nonnegative power-series tail square roots, Perron eigenvector perturbation, and a one-dimensional kernel reduction with uniform parameters. It is substantially larger than the finite rational check and need not block formalizing the numerical obstruction first. No unfinished Lean file is included under the appearance of a completed proof.

24 Further research questions and projects

24.1 Locate the actual fixed-map polyomino saturation constant

Theorem 22.2 proves an obstruction but not its sharp location. A natural next target is a rigorous interval for $\text{spr}(\Phi_X(t, A(t)))$ using upper and lower bounds on the marked profile. Exact lower prefixes give one side; a validated majorant for the remaining marked tail is needed for the other. Determine whether the crossing is interior to the true marked radius, and hence whether Theorem 18.2 applies to this particular map.

24.2 Replace enumeration length by a better recurrence

The 4.3149 obstruction makes a concrete distinction between two research investments: longer exact prefixes versus modifying the geometric recurrence. A structural repair must change the map or its available coefficient inequalities, not merely find a new supersolution of the old corrected map. Identify which local overcounts most affect the Perron crossing, prove a repair, and repeat the certificate audit.

24.3 Resolve the reducible critical case

The example in Section 21 shows why the scalar value $\text{spr}(J) = 1$ is insufficient in a reducible system. Seek a necessary and sufficient block criterion involving residual forcing, reachability of critical blocks, and nonlinear feedback within each block. Distinguish a critical block that is already exact from one receiving arbitrarily small forcing from downstream dependencies. This is a genuine extension beyond the irreducible theorem proved here.

24.4 Classify crossings at the true singularity

Our sharp theorem assumes $r_* < R$. When $r_* = R$, the derivatives of A may diverge and κ need not be finite. Determine how the singular behavior of A , the defect tail, and the Perron gap combine. One should expect different exponents, not automatic reuse of the interior square-root law.

24.5 Obtain second-order truncation asymptotics

The present leading law depends only on κ , β , and η_N . A second-order expansion should involve the complementary inverse $(I - J_*)^{-1}$ on $\ell^T w = 0$, higher derivatives of Φ and A , and the vector direction of the tail. Characterize when oscillating tail directions prevent a single second-order constant even though the leading law remains universal.

24.6 Uniform coefficient asymptotics in two limits

For fixed N , an additional aperiodicity and continuation analysis may turn (56) into a coefficient asymptotic. A more delicate problem lets both n and N tend to infinity. The amplitude shrinks as $\eta_N^{1/4}$ while the radius gap shrinks as $\eta_N^{1/2}$, suggesting a transition when $n\sqrt{\eta_N}$ is of order one. Prove a uniform transfer law and determine its relation to exact prefix agreement for $n \leq N$.

24.7 Certified optimization of the omitted defect

For a fixed data budget, is prefix truncation the best verified correction, or can a sparse collection of rigorously proved higher-degree defects be more effective? The leading sensitivity is Perron-weighted, so unweighted coefficient size is not the right objective. Extend the theory from consecutive prefixes to arbitrary finite certified defect polynomials and characterize the optimal allocation problem.

24.8 Bit complexity of near-critical certificates

The existence of a rational vector follows from strict slack, but the smallest required denominator may deteriorate near the fold. Derive an explicit bit-length bound in terms of the input map, an enclosure of its Perron data, and the separation from ρ_N . Separate the cost of finding a witness from the cost of checking it. The practical integer witness here is an example, not a worst-case complexity theorem.

24.9 Formal marked-profile certification

A verified enumerator or compact enumeration certificate would remove the main inherited computational assumption in the numerical application. The size-18 profile contains much more information than the unmarked polyomino sequence; matching unmarked totals is insufficient. A first formal target could be a smaller prefix and a weaker but nontrivial all-prefix spectral barrier, followed by an incremental extension.

24.10 The distinct neighborhood-refinement question

Bui's question about increasing the geometric neighborhood size remains separate [1]. The fixed-map theorem does not rule out convergence of a sequence of increasingly accurate maps. Formulate conditions on $\Phi^{(k)}$ under which their spectral barriers tend to the true radius, and construct a counterexample when those conditions fail. This would connect the present negative limitation with a positive convergence theory for genuinely improving geometric descriptions.

25 Conclusion

The exact-prefix method has an exact spectral boundary. In an irreducible system, a nonzero uncorrected defect cannot be sustained at or beyond Perron value one, even if every finite-prefix

Jacobian remains below one. This resolves the missing critical case and identifies the limiting radius of the entire hierarchy.

At an interior nonlinear crossing the boundary is quantitative: its distance is asymptotic to a square root of the omitted defect, with an explicit invariant constant. The solution height, scaled branch, and singular amplitude have compatible universal laws. The tail estimate shows why coefficient gaps do not invalidate this conclusion.

For the concrete polyomino map, an exact finite certificate proves a 4.3149 limit on what prefix refinement alone can accomplish. The useful conclusion is methodological: to cross that threshold, one must improve the recurrence information itself, not merely compute a longer initial segment of the same profile. The abstract theorems are proved here; the remaining work is independent validation, formal certification, and the geometric refinement needed to move the spectral barrier.

26 The complete polynomial map used in the certificate

[write] This section is the manuscript’s Appendix A. Its display of the 53 monomials, transcribed from Part I’s code/model.py [18], agrees row for row with (6), and is printed once, there. The manuscript names each row by its lowercase output coordinate $(\Phi_c, \dots, \Phi_z)$, where (6) uses the uppercase type name $(\Phi_C, \dots, \Phi_Z)$; the size variable is ξ , the lowercase z is a profile coordinate, and every coefficient is one.

The five linear partition identities in the supplied profile are

$$f = g + p, \quad g = e + q, \quad h = d + s, \quad r = y + w, \quad t = x + v.$$

The checker verifies these coefficientwise through size 18. The nonnegative defects for the remaining rows are checked as well.

The file data/profiles_18.csv retains the complete input rather than just a floating-point value of $P_{18}(t_0)$. As a result, the reader can reconstruct every exact rational entry of J_{18} directly from the map and the coefficient table. The unmarked total column is retained for provenance, but is not an input to the spectral inequality.

27 A minimal replay algorithm

The numerical core can be expressed as the following exact computation. For each map monomial $\xi^k X_{i_1} \cdots X_{i_m}$, its contribution to $\Phi_X(t_0, a)w$ is

$$t_0^k \sum_{r=1}^m w_{i_r} \prod_{s \neq r} a_{i_s}.$$

Repeated variables occur repeatedly in the sum, giving the correct multiplicity. For example the derivative of ξe^2 contributes $2t_0 a_e w_e$, not merely $t_0 a_e w_e$.

```
t0 = Fraction(10000, 43149)
factor = Fraction(10000001, 10000000)
a = [horner(profile, t0) for profile in profiles]
J = exact_jacobian(t0, a)
image = exact_matrix_vector_product(J, witness)
assert all(image[i] >= factor*witness[i] for i in range(17))
assert all(witness[i] > 0 for i in range(17))
assert strongly_connected(J)
```

The actual checker uses explicit exceptions rather than relying on Python assert, which can be disabled by an optimization flag. The pseudocode above only describes the mathematical test.

The exact arithmetic result is independent of the floating-point approximation used to discover the witness.

For the scalar example, the polynomial sign oracle is

$$H_N(t) = (1 - 3t)^2 - 4t^{N+1}((N - 1) - (N - 2)t).$$

The proved uniqueness of its root in $[1/4, 1/3]$ turns rational bisection into a root certificate. The program checks both endpoint signs again before recording each bracket. It also evaluates the identity

$$H_N(t) = (1 - t)^2(1 - 4t + 4D_N(t))$$

at three independent rational points for every displayed N . Those finite identity checks are safeguards against transcription errors; the all- t identity is the algebraic derivation in Section 20.

28 Source snapshot and evidence ledger

The repository snapshot inspected for this continuation is

74f7f5bdba1aa728b340509701a0fa4e45e8dbf3

The main source directory is

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
polyomino-growth-finite-prefix-corrections/
```

Its `article.tex`, especially Section 8 and the proposed research questions in Section 10, is the mathematical continuation source. Its `code/model.py` supplies the map and local pattern convention; `data/profiles_18.csv` supplies the finite marked counts. The complete paths, blob identifiers returned by the connector, and review boundaries are retained in `02-spectral-saturation-SOURCES.md`. [write] That directory is this report, and its `article.tex` at the pin (blob 4a021fb3) is Part I as printed here: Section 8 is [section 8](#) and Section 10 is [section 10](#). The blobs recorded for `code/model.py` (97b241bd) and `data/profiles_18.csv` (82648328) are also those of the placement commit.

Evidence	What it establishes here
Repository article and README	The exact continuation target and prior results; not blanket verification of the repository.
Repository monomial map	The finite algebraic system used by the checker; its combinatorial interpretation is an inherited source input.
Repository marked CSV	The size-18 coefficients used in the arithmetic certificate; not freshly enumerated here.
Exact checker output	The stated rational inequalities, graph properties, finite consistency identities, and scalar root brackets.
Written proofs	The infinite-level implications, critical classification, spectral limit, and sharp analytic laws.
Primary literature review	Context and attribution for the recurrence, positive-system certificates, Newton theory, and singularity methods; not an exhaustive novelty search.

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- [2] Vladimir Reshetnikov, *ProveIt*, repository snapshot e21766d04c2b8a9b2cd43e563024b3bd1b9. <https://github.com/VladimirReshetnikov/ProveIt>. Repository sources in the next entries are read at this commit.
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- [7] ProveIt, *An exact improvement to $\lambda \leq 4.5235$ for Klarner's constant*, research note dated 12 August 2026, *Combinatorics/Polyominoes/KlarnerConstant/Research/klarner-bound-4.5235.md*. [write] Added in the write; its Sections 1 and 3 give the upper-bound history and the weighted-prefix notation compared in [section 1](#).
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